(№)

2017   ˝Ḥ

ѿ 3

Ꮒ o 7

ৣ ٧ کح غ 10

ضم ΰ کح 15

л کح 20

Έ کح 26

≠ β کح 33

ῆ ‖ Ḥ̂ ΰ کح 38

Ӝ 555 کح ı̋ 40

\/I ˉ ℗ ⅃ 44

ẅ

ẅ¨

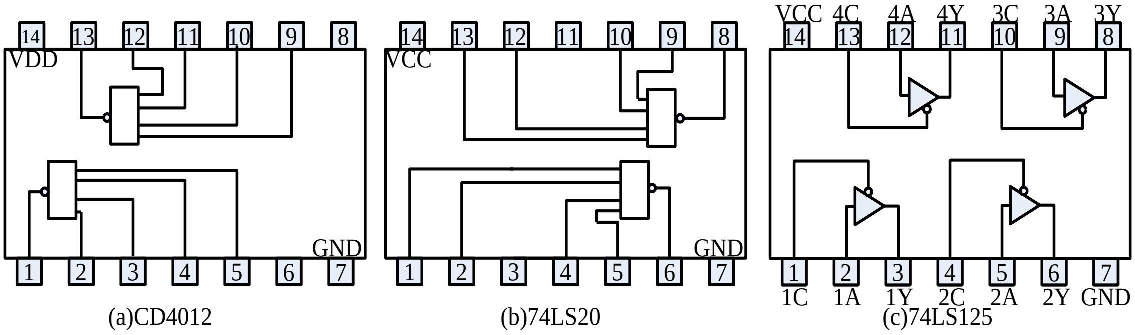
1¨ CMOS ї TTL ẇ Â

2¨ ን ƅ h Â

3¨ ꙰ ẇ ї Â

Ⴎ¨ Л

1¨ ą ꙰ ﺸ CMOS ƅ CD4012˘ TTL ƅ 74LS20 ﻉ꙰ 74LS125 ꞁ ↓ ﯿﺤ 1¨ 1 Â



VCC 4C 4A 4Y 3C 3A 3Y
14 13 12 11 10 9 8
VDD
VCC
GND
GND
1 2 3 4 5 6 7
1C 1A 1Y 2C 2A 2Y GND

(a)CD4012 (b)74LS20 (c)74LS125

ﯿﺤ 1-1 CD4012ȁ 74LS20ȁ 74LS123 ↓

ﯿﺤ 74LS20 ﻍ CD4012 ɥ ﺸ ṗ ﺶ ɥ ɷ ƅ ˘ 74LS125 ﺿ ɷ꙰ Â ꞁ ‶ ⎕ ɥ ↓ ⎕˘ ᛪ ɣ ₒ Ꮞ ˘ ˜ ɽ Ꙓ h ↓ Â ﺸ 7 ﺿ˘ 14 V_{CC} V_{DD} Â Ạ ╠ ﺸ ₒ h ⌠ ꙰ ˘ Ս ˘ CMOS ˜ ƅ Â ╠ Ղ ˘ ɣ ﺸ Ս ﺶ Â

2¨ TTL ƅ h ᾶ

ˆ 1˜ Ŧ

Ŧ Ꮞ ľ 1 Ŀ V_{OH} Â

Ŧ Ꮞ ľ 0 Ŀ V_{OL} Â

ˆ 2˜ ˜

˜ Ꮞ ˜ I_{IH} Â

˜ Ꮞ ᵞ ˜ I_{IL} Â

ˆ 3˜

ƅ ƃ h ǚ Ꮞ Âﺿ ΎṬ₇ Ŧ ƅ Ŧ Ꮞ ᵞ (I_{CCL}) Â

ˆ 4˜

ﺿ ƅ Ŧ Ꮞ ᵞ ˘ ˜ Ḥ˝ ṿ Ꮞ V_{ON}˘ $V_1>V_{ON}$ $V_O=V_{OL}$ Â

ˆ 5˜ ˜

ﺿ ƅ Ŧ Ꮞ 0.9 Ȍ ˘ ˜ Ḥ˝ ṿ Ꮞ V_{OFF}˘ $V1<V_{OFF}$

$V_O=V_{OH}$ Â

ỉ ҇ ẵ ض ƀ ̔P ↓ ҽ˘ ą ˜ ˜ Â

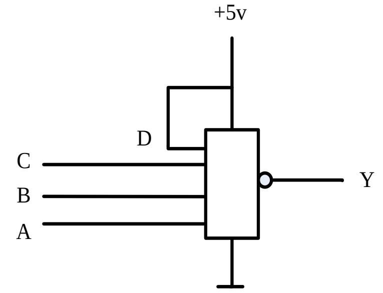
ໍ̈ ῤ ἷ

1˙ CMOS̋ TTL ƀ ẃ

ˆ 1˜ CMOS ƀ ˜ A à B à C №≢ ໍ ɷ ˜ K_0 à K_1 à K_2˘ ˜ D ˘ ₮ Y ⌠ Â ᠉ 1˙ 2 Â 1˙ 1 ↓ ˜ ἷ ΰ ṿ ₮ Â

1˙ 1

˜			₮
A	B	C	Y
0	0	0	
0	1	0	
0	1	1	
1	1	1	

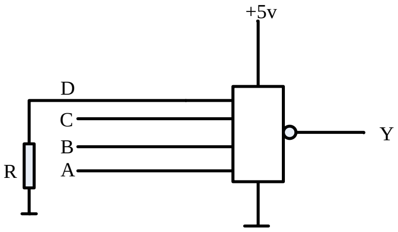


᠉ 1-2 ƀ ẃ

ˆ 2˜ ᠉ 1˙ 2 Ţ ƀ ˜ D R ضُ˘ ỉ ێ ˜ ╠˘ ᠉ 1˙ 3 Â 1˙ 2 №≢ R¸ 30Kẞ ع 200ẞ ˜ ἷ ΰ ṿ ₮ Â

1˙ 2

˜			₮ Y	
A	B	C	R=30Kẞ	R=200ẞ
0	0	0		
0	1	0		
0	1	1		
1	1	1		



᠉ 1-3 ƀ ˜

ˆ 3˜ ᠉ 1˙ 3 Ţ CMOS ƀ CD4012 TTL ƀ LS20 ↓ ƀ ˘ ƀ ˜ Â. 1˙ 2 ₮ Â

ˆ 4˜ ƀ ‖ ˪ƀ Â ᠉ 1˙ 4 Â ݜ Y Â ƀ ẅ ɷ ˜ ‖ Ḥ˜

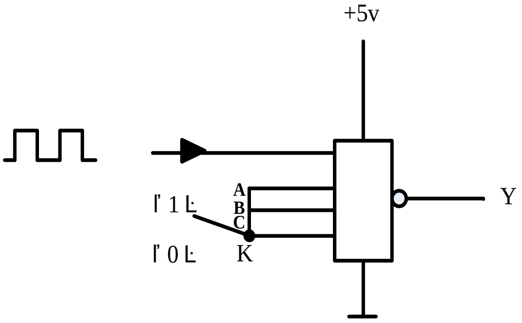
ˆ ӟ 1KHz ‖ ˜˜ ỉ φ ˜

˜ KÂ ˜ K Ɩ' 1 Ŀ ع Ɩ' 0 Ŀ

˘ ˜ à ₮ ɢ 1˙ 3 Â

1˙ 3

˜	K	₮
	1	
	0	

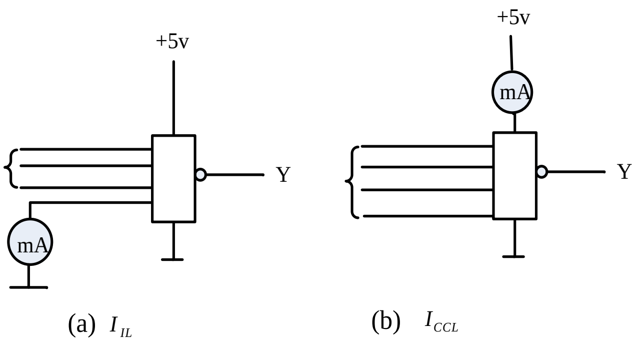

+5v
A
B
C
Ɩ' 1 Ŀ
Ɩ' 0 Ŀ
K
Y

᠉ 1-4 ƀ ‖ Ḥ˜ ˪ƀ

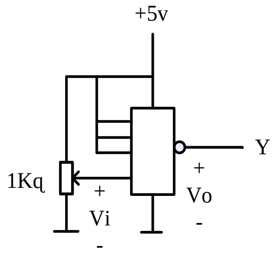
2˙ TTLƀ h ặ

ˆ 1˜ ᵞ ˜ I_{IL} ǵ I_{CCL} Â ﻳﯩ 1˙ 5 Â Â

ˆ 2˜ TTL Â ﻳﯩ 1˙ 6 Â ﺿ 1˙ 4 Ţ Â


+5v
Y
mA
+5v
mA
Y

(a) I_{IL} (b) I_{CCL}


+5v
Y
1Kq
+
Vi
-
+
Vo
-

ﻳﯩ 1-5 ƀ ặ ﻳﯩ 1-6 ƀ

1˙ 4

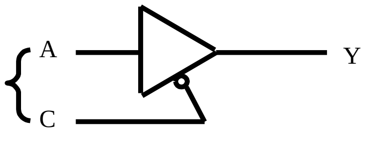
Vᵢ (V)	0.5	1	1.2	1.3	1.4	1.5	1.6	1.8	2	3
Vₒ (V)										

3˙ ᪿ ẇ

ˆ 1˜ ᪿ 74LS125 ˜ ´ ˘ ﻳﯩ 1˙ 7 Â 1˙ 5 ﻼ Â

1˙ 5

C (ᒪ)	A (˜)	Y (Ŧ)
0	0	
0	1	
1	0	
1	1	

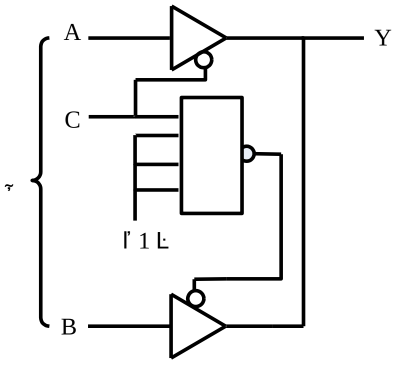


ﻳﯩ 1-7ᪿ ẇ

ˆ 2˜ ᪿ 74LS125 ǵ ƀ 74LS20 ﻳﯩ 1˙ 8 ˘ 1˙ 6 Â

1˙ 6

C	A	B	Y
0	0	0	
	0	1	
	1	0	
	1	1	
1	0	0	
	0	1	
	1	0	
	1	1	



ﻳﯩ 1-8ᪿ ᒪ ẇ

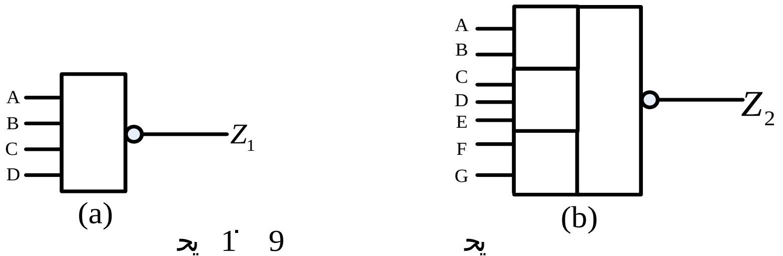
ضمـ̈ ᵱ

1˙ ˘ № Ѣ Ṿ Ղ Â
2˙ شمـ ₮ Ѣ Â
3˙ عجـ Â

ŋ̈

ˆ 1˜ ضيحـ 1˙ 9 Ț˘ Z_1¸ $\overline{AB}$ غ Z_2¸ $\overline{AB+CD}$ ˜ ˘ يحـ Ț

φ ˜ ν ‣


A
B
C
D
Z_1
A
B
C
D
E
F
G
Z_2

(a) (b)

يحـ 1˙ 9 يحـ

ˆ 2˜ Կ Ꞡ TTL Ѣ Ỵ ˜ Ṿ Ọ I_{IL} Ọ ˜ I_{IH} ‣

Ƕ o

ŵ̈

ˆ 1˜ TTL ȁ CMOS ң ɀ Ⱡ ≢ Â

ˆ 2˜ ʋ CMOS ˉ TTL Â

Ƕ̈ ʎ

ض Ț˘ ψ ⌠ TTL ع CMOS ε ˘ ˉ ч ˪ ȁ ʏ ع ˉ Â ˘ ↓ TTL ع CMOS ɦ ᾰ Ⱡ № Â ŵ ᾰ й ع ˜ ΄Y ψ ↓ Ŧ ң ᾰ ˘ ĭ ˜ ɣ Â ɦ ң ɀ Ⱡ ≢ ɣ ĭ CMOS ˉ TTL Â

1˙ ɀ

Ŧ ч ˘ Ŧ Ŧ ˘ ч Â ɢ Ŧ Ѣ ɢ ˘ ɣ ˘ Ŧ ψ ʏ Â Ŧ ʏ ⌠ U ŵ ṿ ʏ ˘ ˜ ψ ǘ Ÿ Â ˘ ṿ Ѣ Â

Ŧ ч ʏ ˘ ˜ Ŧ ˘ ч Â ˜ ˘ Ŧ ʏ ψ Â Ŧ ⌠ U ŵ ʏ ṿ ˘ ˜ ψ ǘ Ÿ Â ˘ ṿ Ӟ Ѣ Â

2˙ TTL ع CMOS o

ˆ 1˜ CMOS ˉ TTL

CMOS ˉ TTL ˘ ɦ CMOS Ŧ TTL ˜ Â TTL ع CMOS ˘ ↕ CMOS ˜ ɣ ˉ TTL Â ɢ TTL ↓ ˜ ʏ ˘ CMOS ↓ Ŧ ʏ ˘ Ѣ ɦ TTL ˪ ˜ ʏ Â ═ ң ɷ' ŵ ŵ ɷ CMOS ą ˘ Ƕ ض CMOS Ŧ ع TTL ˜ ң ˜ CMOS ˉ گ Â

ˆ 2˜ TTL ˉ CMOS

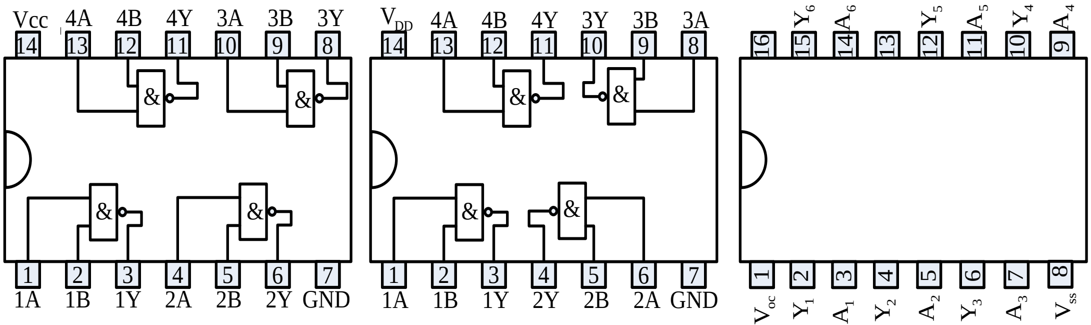
TTL ˉ CMOS ˘ ɦ TTL Ŧ CMOS ˜ Â TTL ع CMOS ˘ ɢ TTL ↓ Ŧ ˘ CMOS ↓ ˜ ˘ ą ң o ʋ Â ч ʋ ═ ɷ ˘ ˜ ɣ ض TTL Ŧ ع ң ŵ ɷ ÂTTL ع CMOS ң o Ӟ˜ ɣ CMOS گ Â

˘ Ѣ ң ˘ Ӟ o ˘ Ḡ ң № ش ƀ˘ Ḡ ˜ Â

3˙ گ Ð W :

74LS00 ч TTL 2 ˜ ضه Ѣ ˘ CC4011 ч CMOS 2 ˜ ضه Ѣ ˘ CC4050 ч Έ

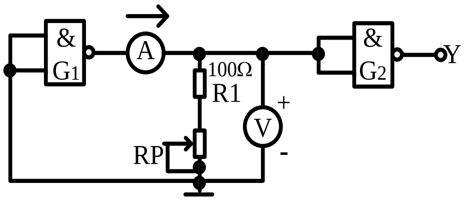
ǁ ݢ Â ٳ ↓ ܪ 2˙ 1 Â


Vcc
4A
4B
4Y
3A
3B
3Y
14
13
12
11
10
9
8
&
1
2
3
4
5
6
7
1A
1B
1Y
2A
2B
2Y
GND
VDD
4A
4B
4Y
3Y
3B
3A
2Y
2A
Y6
A6
Y5
A5
Y4
A4
16
15
Voc
Y1
A1
Y2
A2
Y3
A3
Vss

(a) 74LS00 (b) CC4011 ˏc˘ CC4050

ܪ 2-1 74LS00 ä CC4011 ä CC4050 ↓ ܪ

ᢤ¨ ῤ


&
G1
A
100Ω
R1
&
G2
Y
+
V
-
RP

ܪ 2-2 ʐ

1˙ TTL Ѣ CMOS ʐ

ˆ 1˜ ʐ

ѿ CT74LS00˘ ܪ 2˙ 2

ˆ ܪ Ţ˘ G1 ä G2 ч ѿ ܫ Ţ ɥ ֎

Ѣ ˘ ᵝ ݢ RP ẏ ч 10K ß˜˜

₮ Y ӟ ῧ ῳ ʜ

L˘ U Ӎ Ạ Â

a¨ ᵝ ݢ RP ẏ Â

b¨ ӟ ˘ ˘ ʜ Ѣ ῧ ῳ Â

c¨ ᵝ ݢ RP ẏ˘ ą ῧ ῳ ʜ ↨ ↨ ῧ„ ˘ ᵖ ῧ„ Â

ⅎ V ۼ A ˘ ⅎ ч ₮ ⅎ ẏ U_{OHmin} ۼ ẏ I_{OHmax}˘

˜ 2˙ 1 Ţ Â

d. CC4011˘ a˙ c˘ ˜ 2˙ 1 Ţ Â

2˙ 1 TTL Ѣ CMOS ʐ

ʐ		74LS00	CC4011
ʐ	U_{OHmin} (V)		
	I_{OHmax} (mA)		
ʐ	U_{OLmax} (V)		
	I_{OLmax} (mA)		

ˆ 2˜ ʐ

ѿ 74LS00˘ ܪ 2˙ 3 ˘ ₮ Y ӟ

ῧ ῳ ʜ L˘ U Ӎ Ạ Â

a¨ ᵝ ݢ RP ẏ Â

b¨ ӟ ˘ ˘ ʜ ῧ ῳ Â

c¨ ᵝ ݢ RP ẏ˘ ą ῧ ῳ ʜ ↨ ↨ ˘ ᵖ Â

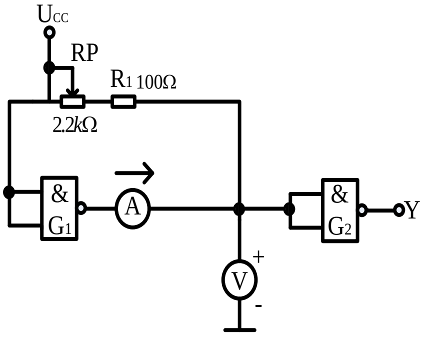
ⅎ V ۼ A ˘ ⅎ ч ₮ ᵞ ⅎ ẏ U_{OLmax} ۼ

ⱱ I_{OLmax}˘ ˜ 2˙ 1 Ţ Â

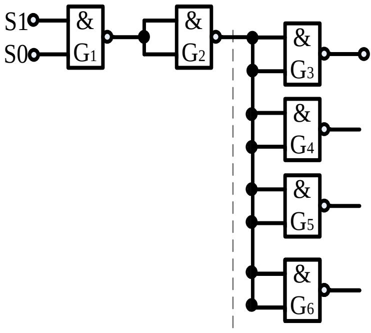
d. CC4011˘ a˙ c˘ ˜ 2˙ 1 Ţ Â

2˙ CMOS ˉ TTL ʐ

ᠴ 2˙ 4 Â ἳ Ţ G_1 ᡎ CC4011 Ţ ŵ ᥨ Ь ’ G_2 ᡎ CC4050 Ţ ŵ ᥨ ǁ ’ G_3˙ G_6 ᡎ 74LS00 Ţ ض ᥨ Ь Â



U_{CC}
RP
R_1 100Ω
2.2kΩ
&
G_1
A
&
G_2
Y
V
+
-



S1
S0
&
G_1
&
G_2
&
G_3
Y
&
G_4
&
G_5
&
G_6

ᠴ 2-3 ʐ

ᠴ 2-4 CMOS ˉ TTL ʐ

ˆ 1˜ ᠴ 2˙ 4 ˘ ἳ Ţ G_1 ɥ ᥨ ˜ S_1 ằ S_2 ɜ ⁛ ˜ K_1 ằ K_2’ G_1 ₮ G_2 ˜ ’ G_2 ₮ G_3 ˜ ˆ ˘ ῤ Ь ˜ G_4 ˙ G_6˜’ G_3 ₮ Y ῢ ῤ Ꮒ L_1 Â

ˆ 2˜ ˜ K_1 ằ K_2 G_1 ɥ ᥨ ˜ ˜ Ḥ˜ ˘ G_1 ằ G_2 ξ G_3 ˜ ᒣ Y¸ S_1 ¥ S_2 Â

ˆ 3˜ ˜ K_1 ằ K_2 ɢ ľ 1 Ŀ ↕ ῢ ῤ Ꮒ L_1 ῢ ῤ Â

ˆ 4˜ G_4˙ G_6 ˜ ᥨ G_2 ₮ ˘ ⌠ L_1 ↨ ˆ L_1 Ь ˘ ˜ ɣ Ṗ ᵥ ŵ ᠱ 74LS00˜˜ ˘ Ṗ S_1 ằ S_2 ξ Y ɥ ᒣ ľ Ь Ŀ ˜ ˘ TTL Ь ᥨ Â

ض¨ ᑭ

ˆ 1˜ № 74LS00 ξ CC4011 ʐ Â

ˆ 2˜ № CC4011 ξ CC4050 TTL ʐ Â

ῧ ४ ﻼ غ

ŵ̈

1̇ ﹰ ۼ ŵ № Â

2̇ ১ ῃ ४ ﻼ àBCḊ 7 ﻼ ɣ ɨ̈ ᠯ r w ɨ̈ ą Â

ʜ̈ /\

ﻼ 3̇ 1

ɣ ɨ̈ ь ῃ ४ ﻼ Â

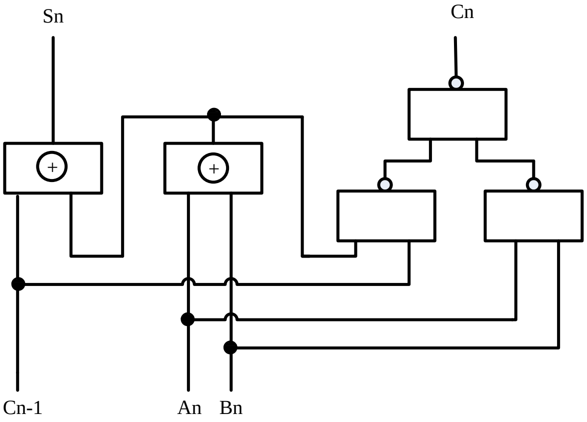
ĩ ﻘﻤ ﹰ ˘ ˜ ɣ ɥ ɷ

ŵ β ʜ ᒪ An ۼ Bn˜ ४˘

ɣ ŵ β β̂ Cn̈ 1˜ Â ₮

β ۼ ԡ Sn˘ h ŵ β β ԡ Cn˘

˜ ɣ ԡ


Sn
Cn
+
+
Cn-1
An
Bn

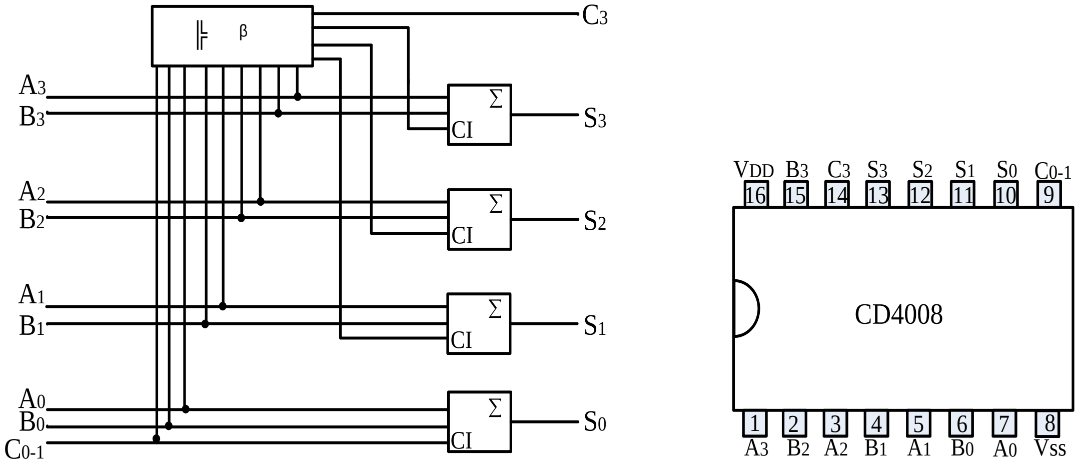
ﻼ 3-1 ῃ ४ ﻼ

$$S_n = A_n \oplus B_n \oplus C_{n-1}$$

$$C_n = (A_n \oplus B_n)C_{n-1} + A_n B_n$$

$$= \overline{\overline{A_n B_n} \bullet \overline{(A_n \oplus B_n)C_{n-1}}}$$

ą ﺸ ˘ ғ7 ŵ β ४˘ ь Â ╞ ą Ţ ῃ ४ ﻼ˜ 7 ᡕ ᡕ˜ β ४ Â C661̂ ɨ̈ ῃ ४ ﻼ˜ ۼ CD4008 ﺿ β ╞ β ῃ ४ ﻼ˜ Âﻼ 3̇ 2 ԡ CD4008 ﻼ ɨ̈ ↓ ﻼ Â ʜ ᒪ $A_3A_2A_1A_0$ ۼ $B_3B_2B_1B_0$ ४˘ C_{0-1} ԡ ɣ β β˘ ₮ ४ ۼ $S_3S_2S_1S_0$ ɨ̈ h β β C_3Â ɨ̇ ʅ ᶍ β β C_0à C_1à C_2ﺿ ᒪ ƀ ﻼ Đ ﺿ ṗ Â

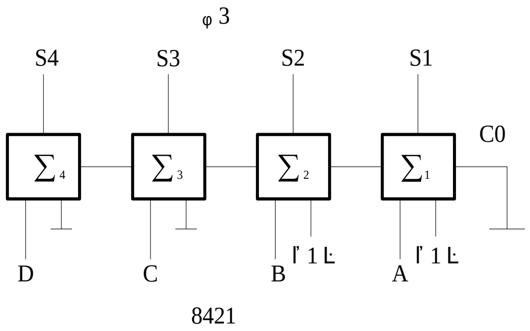

╞ β
C3
A3
B3
Σ
CI
S3
A2
B2
Σ
CI
S2
A1
B1
Σ
CI
S1
A0
B0
C0-1
Σ
CI
S0
VDD
B3
C3
S3
S2
S1
S0
C0-1
16
15
14
13
12
11
10
9
CD4008
1
2
3
4
5
6
7
8
A3
B2
A2
B1
A1
B0
A0
VSS

(a)CD4008 ﻼ

(b)CD4008

ﻼ 3-2

≠ ꞃ ४ ؼ 8421 ʃ φ 3 ˘ ﮨ 3˙ 3 ˘ ˜ ۻ ɷ ꞃ ४ ؼ Âř 1 غ ř 2 ŵ ɷ ˜ 8421 Ŧ ˘ ˈI ₃ ŵ ˜ ľ 1 Ŀ ř 3 ř 4 ŵ ɷ ˜ 8421 Ŧ ₃ ŵ ɷ ˜ ľ 0 Ŀ Ḡ ۺ 8421 Ⅵ Ţ ŵ ≠ ꞃ ४ ؼ ४⁛ 0011 ⊣ ४ 3 ą 8421 ἷ φ 3 ᒪ Â ۻ ᒣ غ

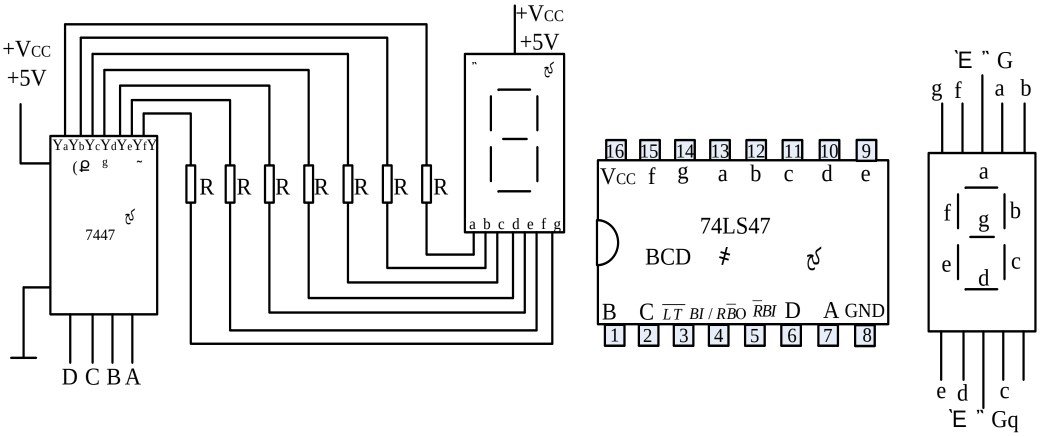

φ 3
S4
S3
S2
S1
C0
∑4
∑3
∑2
∑1
D
C
B
A
ľ 1 Ŀ
ľ 1 Ŀ
8421

ﮨ 3-3 ꞃ ४ ؼ 8421 φ 3

Ţ Ⅵ № Â ϑ غ ˢ ؼ Ð ƀ ἶ ą Â ﮨ 3 ˙ Ꜭ a˜ Ŧ ϑ Ⅵ ﮨ ÂŢ BCD˙ 7 ؼ 74LS47 à CD4511˜ ˜ 8421BCD ABCD ≠ Ŧ a˙ g˘ Ᵽ ≠ Ⅵ ᒪ Â ؼ 74LS47 ᵚ ﮨ 3˙ Ꜭ b˜ Â A à B à C à D ㄩ BCD ˜ ʼ ἶ Ꞁ ᵚ ἇ

ꞩ № P110˙ 111 ˆ ꞕ˜ʼ ą ˘ $\overline{LT}$ à BI/RBO à RBI

ˆ ¯ 5V˜ Â 74LS47 ṗ Ŧ ˩ ؼ˜ ϼ ʐ ɣ ˩ ‟ Â ˆ a˜ ﮨ Ţ R ㄩ Â Ꜧ ᵣ LG5011BSR ᵚ ﮨ 3˙ Ꜭ c˜˜ ㄩ ‟ ˘ G ‟ ¯ 5V Â


+VCC
+5V
Ya Yb Yc Yd Ye Yf Yg
7447
D C B A
R
R
R
R
R
R
R
+VCC
+5V
a b c d e f g
16 15 14 13 12 11 10 9
VCC f g a b c d e
74LS47
BCD ≠ ؼ
B C $\overline{LT}$ $\overline{BI}/\overline{RBO}$ $\overline{RBI}$ D A GND
1 2 3 4 5 6 7 8
G
g f a b
a
f g b
e d c
e d c
Gq

ˆ a˜ ˆ b˜ 7447 ᵚ ﮨ ˆ c˜ LG5011BSR ᵚ ﮨ

ﮨ 3-4

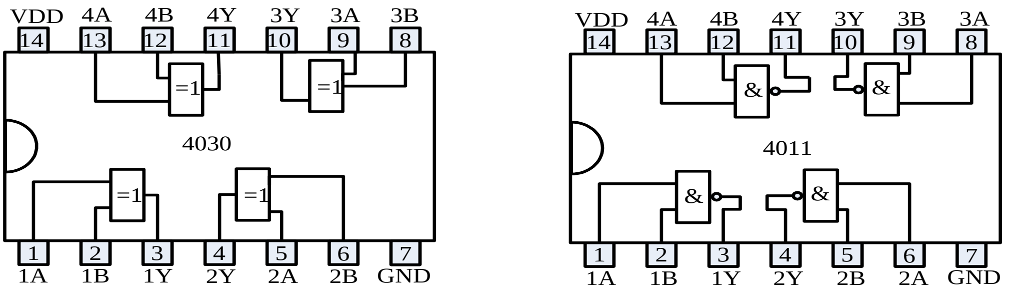
⁞ ̈ ṗ

1˙ ꞃ ४ ؼ

ˆ 1˜ ۺ ဒ ⁛˘ ≠ ˜ ἶ ؼ à ƅ ؼ Ð ᵚ ᒣ Â ﮨ 3˙ 5 ㄩ ۻ CD4030 غ ۻ 2 ˜ ƅ CD4011 ↓ ﮨ Â

ˆ 2˜ ﮨ 3˙ 1 ﮨ Â Ư˘ Â

ˆ 3˜ ꞃ ४ ؼ ᵚ Â 3˙ 1 ƅ ˜ ἷ ˘ ˜ Â


VDD
4A
4B
4Y
3Y
3A
3B
14
13
12
11
10
9
8
=1
=1
4030
=1
=1
1
2
3
4
5
6
7
1A
1B
1Y
2Y
2A
2B
GND
VDD
4A
4B
4Y
3Y
3B
3A
14
13
12
11
10
9
8
&
&
4011
&
&
1
2
3
4
5
6
7
1A
1B
1Y
2Y
2B
2A
GND

ڃ 3-5

3̇ 1 ῃ ৶ ڴ w

An	Bn	C_{n-1}	Sn	Cn
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

2̇ Ț ῃ ৶ ڴ

ˆ 1˜ ڃ 3̇ 2̂ b˜ w ڃ Â ˜ $A_3A_2A_1A_0$ غ $B_3B_2B_1B_0$ №≢ ⌠ ԡ ˜ Ț Â ₮ $S_3S_2S_1S_0$ ἳ C_3 ᵝ №≢ ⌠ Ꮢ ⱷ ڴ Ț Â ᵞ ᵝ C_{0-1} ڞ˜ U˜ Â

ˆ 2˜ Йџ ᒪ ৶ ˘ ˜ 3̇ 2 Ț Â

ˆ 0010˜ $\overline{2}$ ˆ 0101˜ 2‚ ‚

ˆ 0100˜ $\overline{2}$ ˆ 0011˜ 2‚ ‚

ˆ 1010˜ $\overline{2}$ ˆ 0101˜ 2‚ ‚

3̇ 2 ῃ ৶ ڴ

A_3 A_2 A_1 A_0	B_3 B_2 B_1 B_0	C_3	S_3 S_2 S_1 S_0	ⱴ ᒪ
0 0 1 0	0 1 0 1			
0 1 0 0	0 0 1 1			
1 0 1 0	0 1 0 1			

3̇ ڃ 3̇ 3 ˘ 8421 A ầ B ầ C ầ D ˆ ˜ ˜ ˜

φ 3 ˘ ₮ $S_4S_3S_2S_1$ №≢ ⌠ ڞ ⱷ ڴ Ț˘ ˜ 3̇ 3 Ț Â

3̇ 3 8421 φ 3

8421	φ 3
D C B A	S_4 S_3 S_2 S_1
0 0 0 0	
0 0 0 1	
0 0 1 0	
0 0 1 1	
0 1 0 0	
0 1 0 1	
0 1 1 0	

0 1 1 1 1 0 0 0 1 0 0 1	

4˙ ˙

ˆ 1˜ ﻳﺢ 3̇ 4̇ a˜ Â ˈ ɢ ˘ ɣ ≢ Â

ﺿﻲ ȁ ҥ Ʊ˘ ˜̃ Â

ˆ 2˜ ≠ ӟ ҉ ˜̃ K_0˙ K_3 ﻛﺢ ˜ 3̇ 4 Ț 8421BCD ˘

Â

3̇ 4 ˙

$\overline{LT}$	D C B A	ﻘﻤﻢ
1	0 0 0 0	
1	0 0 0 1	
1	0 0 1 0	
1	0 0 1 1	
1	0 1 0 0	
1	0 1 0 1	
1	0 1 1 0	
1	0 1 1 1	
1	1 0 0 0	
1	1 0 0 1	
1	1 0 1 0	
1	1 0 1 1	
1	1 1 0 0	
1	1 1 0 1	
1	1 1 1 0	
1	1 1 1 1	
0	¦ ¦ ¦ ¦	

*5˙

ɣ҉ ɤ Ѣ ˘ ˙ ҥ ɷ ﺿﻤ ᵝ Ҥ ᒪ

ɤ ˆ ﺿﻲ Ⅵ ɣ ṗ ˜ Â ﺿﻲ Ạ Ț ŋ ɤ ﻛﺢ ˘ ˜̃

ﺿﻲ ӟ Ӥ ѿ ﺸﺠ ˊ ҉ Â ﺿﻲ ҉ ѿ ﺸﺠ ˊ ҉ Â ą

ɷ ˘ Ѣ Ӱ Â Ӥˈ

ˆ 1˜ Ț ŋ ɤ ﻛﺢ ɛ́ Ŧ $S_3S_2S_1S_0$ ˜ ˙ ˘ ה Â

ˆ 2˜ ṗ 2 Ț ﯾ ˘ Â

ﺿﻤ̈ ɚ

№ Â ﺠﻊ

л̈ ɜ̈ ѣ

1̈ ɜ̈

ˆ 1˜ ɜ̈ ῃ ѵ ݣ ï̋ ˙ ƀ /| Â

ˆ 2˜ ϩ Ҭ ῃ ѵ ݣ à ݣ ï̋ ν r ï̋ һ ᾅ ȁ ą
Һ Â

2˙

ˆ 1˜ v ݜ 3˙ 1 ῃ ѵ ݣ Ӥ ˜ ˈ

ŵ $F = A \oplus B$

Ŷ $F = \overline{A}$

ˆ 2˜ ≠ CD4008 ԣ ɶ ῃ̃ β Һ └ ѵ

ˆ 3˜ ضي ῤ 5 Ҭ˘ Ӥ ѵ ˈˆ 1101˜ $\overline{}_2$ ˆ 1010˜ $_2$ ἑˆ 1101˜ $\overline{}_2$ ˆ 1110˜ $_2$ ˘
˙ Ʉ ч Ʉ

ضم ǖ ݣ

ŵ̈

1̈ ఌ ǁ ǖ ƀ Â

2̈ ǖ ݣ ẅ Â

3̈ ݵ ǖ ݣ ε Â

ҤÜ ʎ

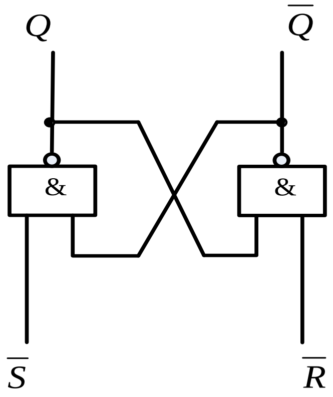
ǖ ݣ ˘ ỉ ẅ № Ṙ S ǖ ݣ à J K ǖ ݣ à D ǖ ݣ غ T ǖ ݣ Â

1̈ RS ǖ ݣ

RS ǖ ݣ Ⴈ ǖ ݣ Â ɥ ɷ ˊ ï ̈ Â ą 74LS00 ضمقم 2 ˜ ƅ Ţ ɥ ɷ Ḃ̃ Â ݴ 4̇ 1 Â

RS ǖ ݣ

$$\begin{cases} Q^{n+1} = S + \overline{R}Q^{n} \\ RS = 0\,(\qquad \text{Đ}) \end{cases}$$



Q
$\overline{Q}$
&
&
$\overline{S}$
$\overline{R}$

ݴ 4-1 R-S ǖ ݣ

2̈ J K ǖ ݣ

J K ǖ ݣ h ǖ ݣ ɥ ŵ̆ ẅ ˎ˘ ą à Âݴ 4̇ 2

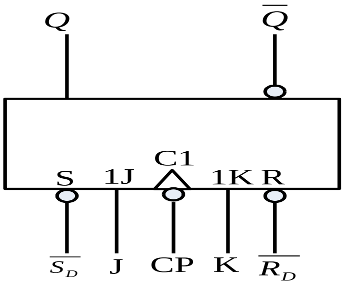
J K ǖ ݣ ˜ Âݴ Ţ R à S Ⴈ $\overline{R_D}$ à $\overline{S_D}$ ˜ ԿԿ β à β̂ Ⴈ ľ 0 Ŀà ľ 1 Ŀ̃ ˜ ˘ ݬ ˘ γ ˩ Â $\overline{R_D}$, 0 $\overline{S_D}$, 0˜ ˘ ǖ ݣ ƅ ỉ CP ï̃ Ł ˜ ڗ ÂCP ԿԿ

˜ ˘ $\overline{S_D}$, $\overline{R_D}$, 1 ˘ ǖ ݣ ضي CP ǁ ƀ Ӎ ǖ ÂCP ݬ

ǖ ݣ ǖ ǖ ݣ ÂJ à K ԿԿ ǖ ݣ Ł Ḥ̃ ˜ ˘ J à K ɷ ˜ ˘ ꟷ ε ԿԿ ľ ƅ Ŀ ˜ ÂQ à $\overline{Q}$ ԿԿ ɥ ɷ ε Ŧ Â74LS76 TTL ضم ï̃ Ӎ J K ǖ ݣ Â

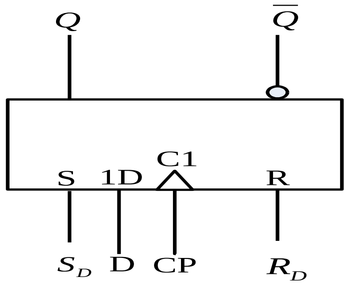
J K ǖ ݣ ɢ à № à ǁ ǖ Ţ˘ ˈ

$$Q^{n+1} = J\overline{Q^{n}} + \overline{K}Q^{n}$$



Q
$\overline{Q}$
C1
S
1J
1K
R
$\overline{S_D}$
J
CP
K
$\overline{R_D}$

ݴ 4-2 J-K ǖ ݣ ˜



Q
$\overline{Q}$
C1
S
1D
R
S_D
D
CP
R_D

ݴ 4-3 D ǖ ݣ ˜

3˙ D ΰ ك

D ΰ ك ѿ ΰ ك Â 74LS74 Ţ ӊ ҩ ε ҉ ˥ D ΰ ك˜

˜ ܲ 4˙ 3 Â

D ΰ ك ɢ à ᒪ Ţ˘ ᵝ à ع №

˦ ˝Ḫ

Q^{n+1} ˛ D

ჲ ¨ ṗ ȉ

1˙ RS ΰ ك

RS ΰ ك˘ ˜ $\overline{R}$ à $\overline{S}$ №≢ ӟ ˜˘ ₮ Q à

$\overline{Q}$ №≢ ك Â ȋ ˜ ˘ ɢ 4˙ 1 Â (ч

ض 2 ˜ ƅ 74LS00 Â)

4˙ 1

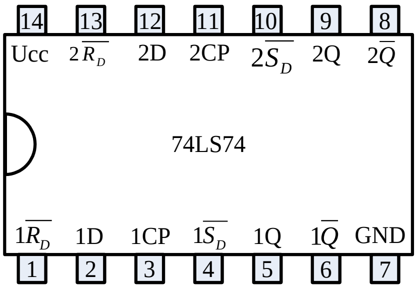
$\overline{R}$	$\overline{S}$	Q	$\overline{Q}$	₮
1	1			
1	0			
0	1			
0	0			

2˙ D ΰ ك

ȋ D ΰ ك 74LS74˙ ȉ ᴡ ܲ 4˙ 4 Â

ˆ 1˜ ᵝ $\overline{S_D}$ ع ᵝ $\overline{R_D}$ ᴡ

D à $\overline{S_D}$ à $\overline{R_D}$ №≢ ˜ K₀ à K₁ à K₂˘ CP ˦ ǁ ₮˘ Q à $\overline{Q}$ ك L₁ à L₂ Â 4˙ 2 ˘ ض $\overline{S_D}$ à $\overline{R_D}$ ƌ ȋ D ع CP ˘ $\overline{S_D}$ ع $\overline{R_D}$ ₮ ᒪ ƌ Â


14
13
12
11
10
9
8
Ucc
$2\overline{R_D}$
2D
2CP
$2\overline{S_D}$
2Q
$2\overline{Q}$
74LS74
$1\overline{R_D}$
1D
1CP
$1\overline{S_D}$
1Q
$1\overline{Q}$
GND
1
2
3
4
5
6
7

ܲ 4-4 74LS74 ܲ

4˙ 2

D	CP	$\overline{S_D}$	$\overline{R_D}$	Q	$\overline{Q}$
¦	¦	0	1		
¦	¦	1	0		

ˆ 2˜ D ΰ ك ᴡ

ȋ D ˘ $\overline{S_D}$ ع $\overline{R_D}$ ΰ ك ᵝ ᵝ Â 4˙ 3 ˘

ἰ ẇ ˘ ﺿﻲ Ţ Â

4˙ 3

D	CP	Q^n	Q^{n+1}
0	Ŕ	0	
		1	
	Ṛ	0	
		1	
1	Ŕ	0	
		1	
	Ṛ	0	
		1	

ˆ 3˜ D ῢ ﮏ T︶ ῢ ﮏ

D ˜ ﻍ $\overline{Q}$ T︶ ῢ ﮏ˜ CP ӟ 1KHz ǁ˘ ӈ

ﮏ CP ﻍ Q Â

3˙ JK ῢ ﮏ

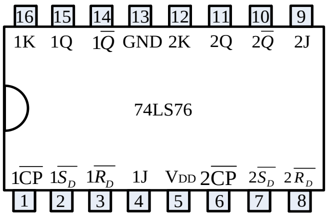
ἵ JK ῢ ﮏ 74LS76˘ ἰ ẇ ﻳﮧ 4˙ 5 Â

ˆ 1˜ J à K ﻍ $\overline{S_D}$ à $\overline{R_D}$ №≢ ˜˘ CP ˧ ǁ ₮ ˘ Q à $\overline{Q}$

Â 4˙ 4 ˘ $\overline{S_D}$ à $\overline{R_D}$ ₮ └ ђ Â

4˙ 4

CP	J	K	$\overline{S_D}$	$\overline{R_D}$	Q	$\overline{Q}$
¦	¦	¦	1	0		
¦	¦	¦	0	1		


16 15 14 13 12 11 10 9
1K 1Q 1$\overline{Q}$ GND 2K 2Q 2$\overline{Q}$ 2J
74LS76
1$\overline{CP}$ 1$\overline{S_D}$ 1$\overline{R_D}$ 1J VDD 2$\overline{CP}$ 2$\overline{S_D}$ 2$\overline{R_D}$
1 2 3 4 5 6 7 8

ﻳﮧ 4-5 74LS76 ﻳﮧ

ˆ 2˜ JK ῢ ﮏ ẇ

ἵ J à K ˘ $\overline{S_D}$ ﻍ $\overline{R_D}$ ῢ ﮏ β β Â 4˙ 5 ˘

ἰ ẇ ˘ ﺿﻲ Ţ Â

ˆ 3˜ JK ῢ ﮏ T︶ ῢ ﮏ

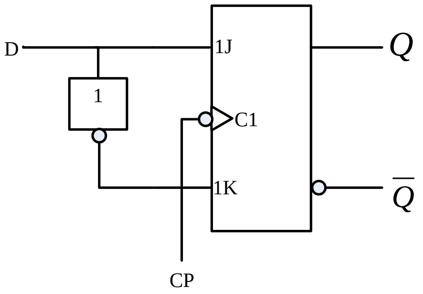
J à K ľ 1 Ŀ T︶ ῢ ﮏ Â CP ˜ 1KHz ǁ˘ ἵ

ﮏ CP ﻍ Q Â

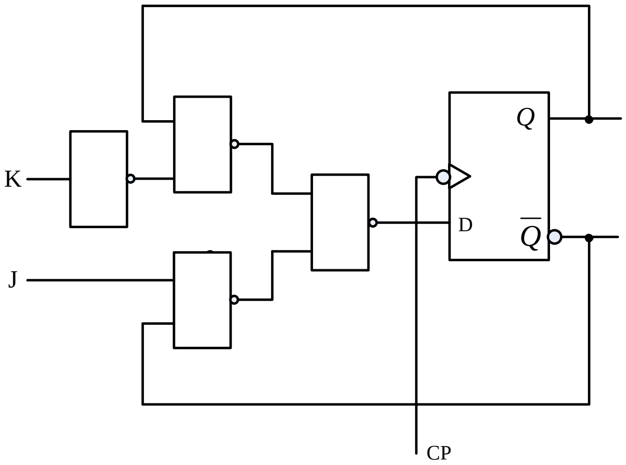
*4˙ D ῢ ﮏ ﻍ JK ῢ ﮏ ε

ˆ 1˜ JK ῢ ﮏ D ῢ ﮏ˜ ἰ ẇ Â ἇ ﻳﮧ 4˙ 6 Â

ˆ 2˜ D ῢ ﮏ JK ῢ ﮏ˜ ἰ ẇ Â ἇ ﻳﮧ 4˙ 7 Â


D
1
1J
C1
1K
CP
Q
$\overline{Q}$

ﯾﺪ 4-6 JKŸ D ﯾﺪ


K
J
Q
D
$\overline{Q}$
CP

ﯾﺪ 4-7 DŸ JK ﯾﺪ

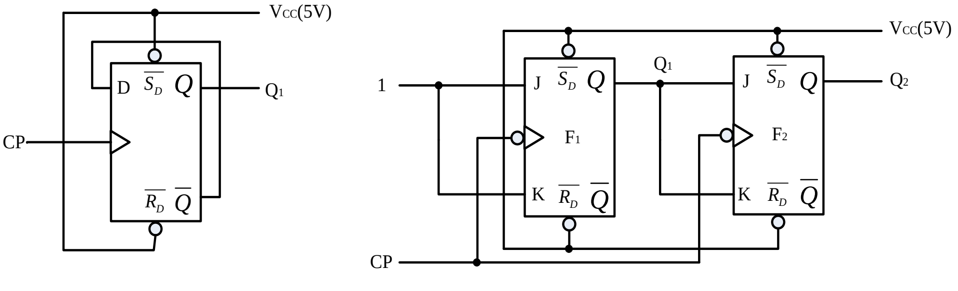
4˙ 5

J	K	CP	Q^n	Q^{n+1}
0	0	Ŕ	0	
			1	
		Ŗ	0	
			1	
0	1	Ŕ	0	
			1	
		Ŗ	0	
			1	
1	0	Ŕ	0	
			1	
		Ŗ	0	
			1	
1	1	Ŕ	0	
			1	
		Ŗ	0	
			1	

5˙ ΰ ﻛﺞ

ˆ 1˜ ҥ№ ầﻀﻣ№ ʬ Â

74LS74 ﻍ 74LS76 №≢ ﯾﺪ 4˙ 8 ˘ CP ǁ˘ ỉ̋ ﻛﺞ №≢


Vcc(5V)
D
$\overline{S_D}$
Q
Q1
CP
$\overline{R_D}$
$\overline{Q}$
Vcc(5V)
1
J
$\overline{S_D}$
Q
Q1
F1
K
$\overline{R_D}$
$\overline{Q}$
J
$\overline{S_D}$
Q
Q2
F2
K
$\overline{R_D}$
$\overline{Q}$
CP

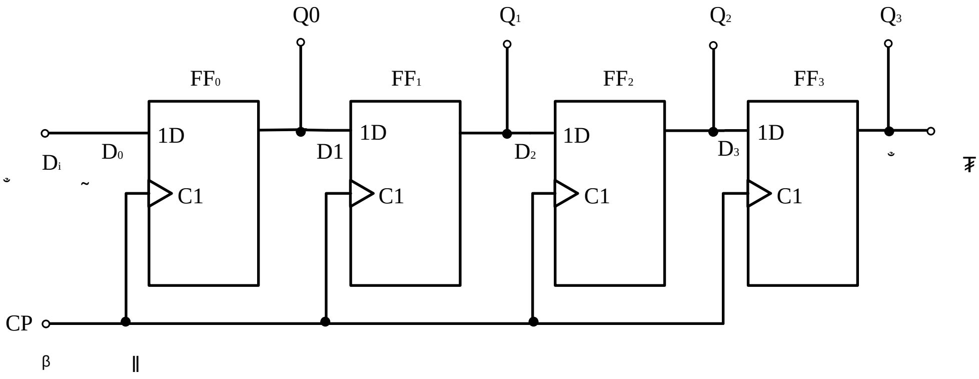
ﯾﺪ 4- 8

Q_1 ƅ CP غ Q_1Q_2 ƅ CP ˜̃ ˘ شمضي ⁛ ₮ ỉ ˘ ẇ Â
ˆ 2˜ β كح

74LS74 يح 4̇ 9 ˘ ˜ CP ᚁ ǁ˘ ₮ $Q_0 \sim Q_3$ №≢ ˘

D_i غ ضم ɷ D ǖ كح ˜ $\overline{S_D}$ شخ ľ 1 Ŀ ضم ɷ D ǖ كح ˜ $\overline{R_D}$ ˜̃

K˘ K β ʿP ľ 1 ĿÂ ضي CP ƀ Ӆ˘ ǖ كح ₮ ˘ 4̇ 6˘
ẇ Â


Q0
Q₁
Q₂
Q₃
FF₀
FF₁
FF₂
FF₃
1D
1D
1D
1D
C1
C1
C1
C1
Dᵢ
D₀
D1
D₂
D₃
CP
₮

يح 4-9

4̇ 6

D_i CP	Q_0^n Q_1^n Q_2^n Q_3^n	Q_0^{n+1} Q_1^{n+1} Q_2^{n+1} Q_3^{n+1}
1 Ŕ		
1 Ŕ		
1 Ŕ		
1 Ŕ		

ضم̈ ҏ

˘ № Âعج Â

Ԓ̈ ӟ ȉ̋

1̇ ӟ
ӟ ʵ ǖ كح ẇ ȁ ǖ ȉ̋ ṗ Â
2̇
RS ǖ كح Ӌ̦ Ʉ ƅ ▸ ᵥ ῷ ₮ ƅ ▸

Ј ݣ

ŵ¨

1¨ Ţ ݣ ƀ / ƅ w Â

2¨ ኄ ݣ Â

Ю¨ /

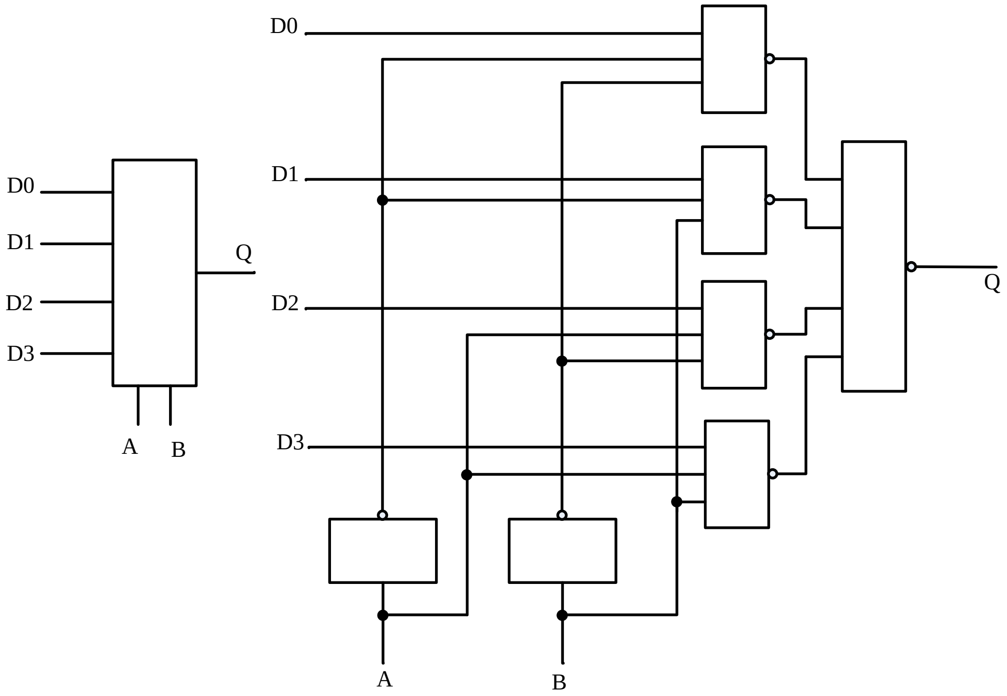
ݣ ﻴ Ð Ḩ ŵ Â ڮ ˜ D_0 à D_1 ˙ ɷ └ ˜ A à B ˙ ع ŵ ɷ ₮ Q Â ﺿﻲ └ ∛ Ḥ ˇ ˇ Ḥ ₮ ˇ ⌠ ₮ Â ﺮﻳ 5¨ 1 ŵ ɷ ƅ ﺿﻪ ŵ ݣ ﺮﻳ ƅ Â

D_0 à D_1 à D_2 ع D_3 ﺿﻪ ɷ ˜ ˇ ˜ ɣ ˜ ﺿﻪ Ḥ ˇ └ ݣ A ع B ∛ Ḥ ﻴ ŵ ˆ ᴅ A¸ 0˜ B¸ 0˜ ˇ ƅ ˇ ∛ ⌠ Ḥ ˆ ᴅ D_0˜ ₮ Â ч

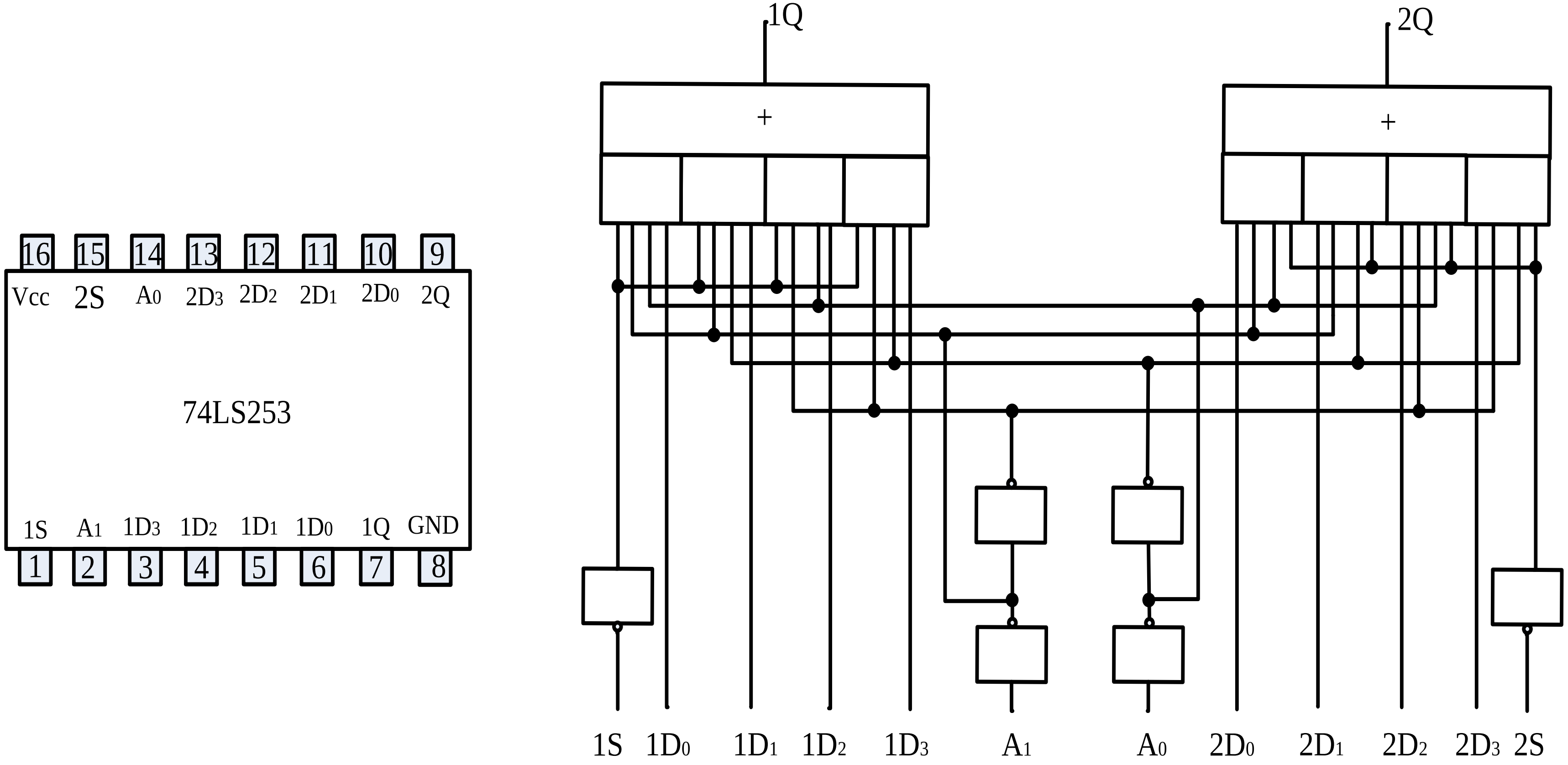
$$Q=\overline{A}\,\overline{B}D_0+\overline{A}BD_1+A\overline{B}D_2+ABD_3$$

ą З ɣ ﺿﻲ └ ˜ ∛ ŵ Ю └ Ḥ ˇ ą ₮ ŵ ɣ Ḥ ˇ ˇ ɣ З ŵ ˜ Ð Â

Ţ L253 ŵ ɷ ĩ ﺿﻪ ŵ ݣ ĩ ﺮﻳ ƅ ↓ ﺮﻳ ﺮﻳ 5-2 ĩ Ţ ˜ ˇ 1S à 2S ч ˇ A_0 à A_1 ч └ ݣ └ Ḩ ɷ ﺿﻪ ŵ ݣ ƀ Â L253 w 5-1 ˇ ĩ Ţ Ø Â S¸ 1 ˇ A_0 à A_1 G¸ Ƕ ˇ ݣ ₮ Q G 0˜ Ḥ ƀ ˆ S ¸ 0˜ ݣ ƀ Â



ﺮﻳ 5-1 ƅ ﺿﻪ ŵ ݣ


16
15
14
13
12
11
10
9
Vcc
2S
A0
2D3
2D2
2D1
2D0
2Q
74LS253
1S
A1
1D3
1D2
1D1
1D0
1Q
GND
1
2
3
4
5
6
7
8
1Q
2Q
+
+
1S
1D0
1D1
1D2
1D3
A1
A0
2D0
2D1
2D2
2D3
2S

يد 5-2 74LS253 ĭ ضم W̄ كج

5˙ 1 L253 ẅ

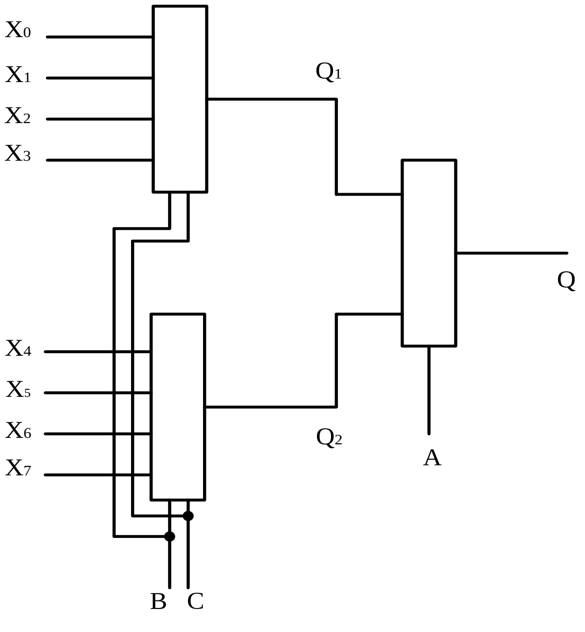
˜			₮
A_1	A_0	S	
Φ	Φ	1	
0	0	0	
0	1	0	
1	0	0	
1	1	0	

ƀ ﻘﻤ ƌ ˜ ᴸ ƀ ˘ ₮ ɛ Ḥ ˆ ⱦ /| ƀ ɨ ˜˜ ₮ ƀ ₮ Â ﻙ ᴸ ɢ ˜ ﺿﺮﻴﺨ ẅ ɷ ﻀﻣ ẅ ﻙ ﻀﻣ ɷ ﺿﺮﻴﺨ ˘ ɥ ɷ ᴸ ˜ ' ŋ̃ ẅ ﻙ ŋ̃ ɷ ﺿﺮﻴﺨ ˘ ɣ ɷ ᴸ ˜ ˘ ɨ φ Â ᴸ Ḥ Ţ ẅ ﺿﺮﻴﺨ ˘ ﺿﺮﻴﺨ ˜ Ḥ ₮ Â ﻙ ˘ Ӎ . ɐ ₃ ᴰ '

1˙ Ḥ Ȅ ẅ ɷ ˆ ˜ ᴰ

Ḥ ˜ ﻙ ɣ̀ ˜ ˘ ᴸ Ḥ ˘ ą Ţ ẅ Ḥ ˜ ˜ Ȅ " ˆ ˜˜ Ḥ Â Ḥ ˘ ˜ ɣ ɷ

ﻙ ﹻ ą ˘ ﺱ 5˙ 3 ɥ ɷ ﻀﻣ ẅ ƀ ẅ ɷ Ⱨ ẅ ﹻ ŋ̃ ẅ ˘ ᴸ A à B à C ɷ Â



ﺱ 5-3 ﻙ ﹻ ą

2˙ ɨ ɥ ɣ ɨ ﻙ Ḥ ˜ ˘ ą ƌ ᴸ Ḥ ẅ ɰ ɨ ˘ ↕ ﺿ ₮ ˜ ɣ ₮ Â

3˙ ₱

ˆ 1˜ ŋ �v ﻙ ŋ ɣ ﻙ ƌ ɥ ẅ˘ ἐ S_i ɨ h β β C_i ɥ

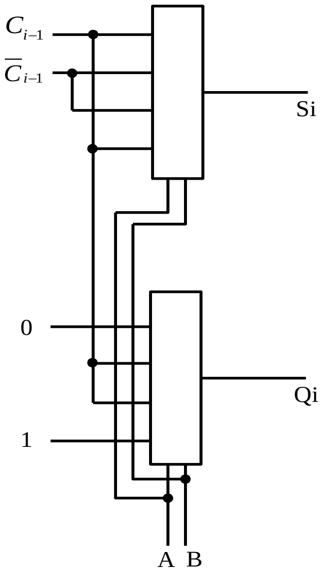
$$S_i=\bar{A}\bar{B}C_{i-1}+\bar{A}B\bar{C}_{i-1}+A\bar{B}\bar{C}_{i-1}+ABC_{i-1}$$

$$C_i=\bar{A}BC_{i-1}+A\bar{B}C_{i-1}+AB\bar{C}_{i-1}+ABC_{i-1}$$

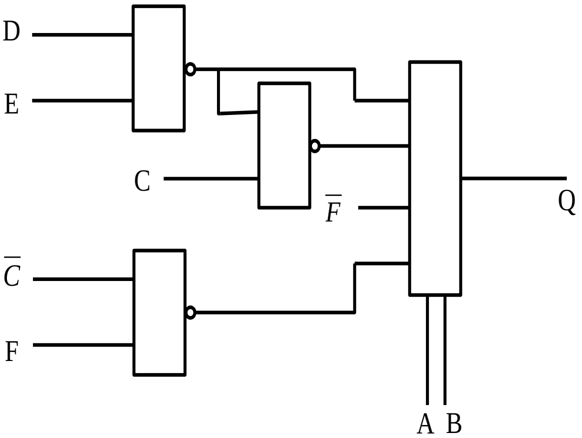
˜ ą Ғ ŋ ɣ ﻙ ƌ ˘ 3˜ ɣ ẅ ɷ ɨ ﻀﻣ ẅ ﻙ Â ﺱ 5˙ 4˜ ˘ ₇ ɨ Ţ ɥ ɷ ɨ ˆ A à B˜ ƀ ɥ ᴸ Ḥ ˘ ₃ ẅ ɷ ɨ ˆ C_{i-1}˜

ए ݣ ݹ ˜ ˘ ܪ ҉ ŋ ए ݣ Â
ˆ 2˜ ɨ́ ₱

ݎ 5ˆ 4ˆ ˘ ŵ ɷ ض ŵ ݣ ҉ ɨ́ ₱ ˘ ᵖ ₮˘
ɣ ₱ ˘ ₱ ԋ غ ˘ ݏ ₱ . ђ Â
щ ˘ ŋ̃ ŵ ݣ ɣ ض ɨ́ ₱ Â


C_{i-1}
$\overline{C}_{i-1}$
Si
0
Qi
1
A B

ݎ 5-4 ݣ ŋ ए ݣ


D
E
C
$\overline{F}$
$\overline{C}$
F
Q
A B

ݎ 5-5 ݣ ₱

$$Q=\overline{A}\,\overline{B}\,\overline{D}+\overline{A}\,\overline{B}\,\overline{E}+\overline{A}B\overline{C}+\overline{A}BDE+A\overline{B}F+ABC+AB\overline{F}$$

Ţ ɨ́ ˘ ˆ ɣ ݣ ˘ ӟ ɣ
ݣ ៈ ɨ̀ ז Â ᵖ ˘ ₱

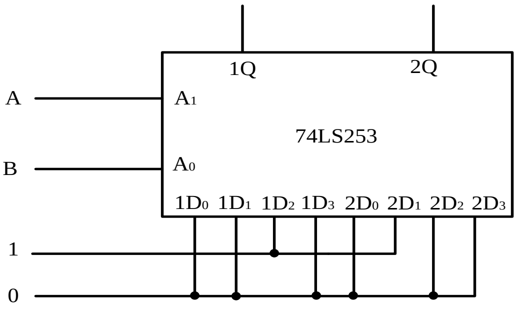
$$Q=\overline{A}\,\overline{B}\,\overline{D}+\overline{A}\,\overline{B}\,\overline{E}+\overline{A}B\overline{C}+\overline{A}BDE+A\overline{B}F+ABC+AB\overline{F}$$

ɾ ₱ ˆ ɣ ₮˘ ݹ ĕ ݜ´ ԑ A غ B ԋ ɷ ɨ́ ˘ ˆ

$$Q=\overline{A}\,\overline{B}(\overline{D}+\overline{E})+\overline{A}B(\overline{C}+DE)+A\overline{B}F+AB(C+\overline{F})$$
$$=\overline{A}\,\overline{B}\,\overline{DE}+\overline{A}B\overline{C¥\overline{DE}}+A\overline{B}F+AB\overline{\overline{C}F}$$

ˆ ݎ 5-5 Â

4ˆ


1Q
2Q
A
A_1
74LS253
B
A_0
$1D_0$ $1D_1$ $1D_2$ $1D_3$ $2D_0$ $2D_1$ $2D_2$ $2D_3$
1
0

ݎ 5-6 ݣ

ݎ 5ˆ 6 ˘ ԋ ɷ A ƅ B №≢ ض ŵ ݣ └ A_0 ƅ A_1
˜ʾ A>B ˘ 1Q ₮ 1ʼ A<B ˘ 2Q ₮ 1ʼ A¸ B ˘ 1Q ƅ 2Q ₮ ݜ ч 0 Â

҉ ¨ ╠‰

1¨ ఌ Ţ ݣ /ˈ à w ƅ ą Â

2˙ ₮ ĸ Ⱶ 2 ৯ Â

3˙ № ĸ Ⱶ 3 ƅ ĸ Ⱶ 4 Ţ ẁ Â

ض̈ ą ĸ Ⱶ ƅ

1˙ 74LS253 ˆ 5˙ 1 ˜

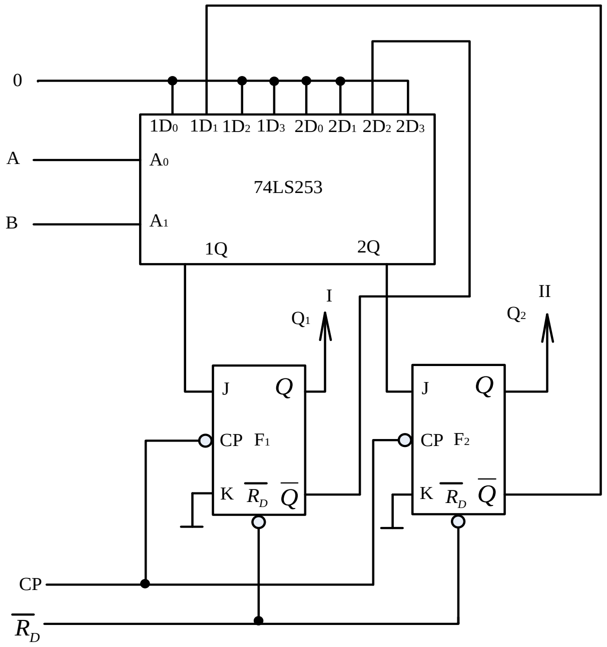
2˙ ض ŵ ɗ ₱

$$Q=\overline{A}\,\overline{C}\,\overline{D}+A\overline{C}DE+AC\overline{D}+\overline{B}CD+BC\overline{D}F+\overline{C}D\overline{E}+\overline{C}DF+CD\overline{F}$$

Â

3˙ ƀ ɏ ɗˆ β ˜

ą ৯ 5˙ 7 Â ȕ ɗ ₮ Q_1 ƅ Q_2 ȕ ῷ ҥ Â ῷ ȕ ɗ 0Â



0
A
B
1D0 1D1 1D2 1D3 2D0 2D1 2D2 2D3
A0
A1
74LS253
1Q
2Q
I
II
Q1
Q2
J
Q
CP
F1
K
RD
Q
J
Q
CP
F2
K
RD
Q
CP
RD

৯ 5-7 ɏ ɗˆ ɾ β ˜

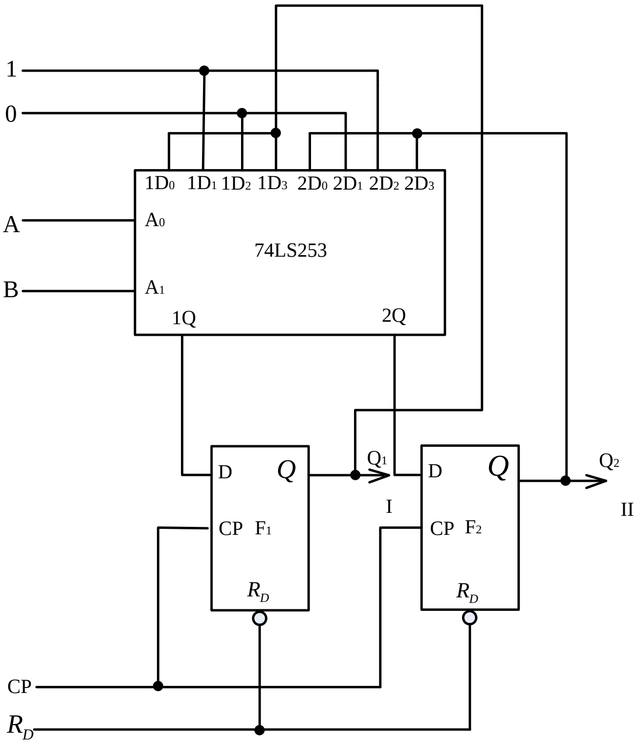
№≢ ɥ ɷ A ɛ̇ B̑ ɥ ɷ β ˜ β ɏ ˜ A_1 ƅ A_0 ˘ β Â A>B˘ ↕ Q_1˛ 1˘ I˶ ˘ A<B˘ ↕ Q_2˛ 1˘ ň˶ Â ˝ ɣ ĸ ஂ Έ β ҥ ˜ 5˙ 2 Ţ˘ Ư ɢ Ţ Â

5˙ 2 ĸ Ⱶ 3

	ŵ						ҥ						ஂ					
A																		
B																		
Q_1																		
Q_2																		

*4˙ ƀ ɏ ɗˆ ɣ β ˜

৯ 5˙ 8 ˘ ĸ Ⱶ 3Â


1
0
1D0
1D1
1D2
1D3
2D0
2D1
2D2
2D3
A
A0
B
A1
74LS253
1Q
2Q
D
Q
CP
F1
RD
Q1
I
D
Q
CP
F2
RD
Q2
II
CP
RD

يج 5-8 Ɏ ݣ ɾ ˠ ᵝ ˜

ꞁ̈ ɘ

Ŧ ˘ Ώ Ŧ ˘ Â

Έ̇

1̈ ضم ѿ ݣ ꟿ Έ ѿ ˒

2̇ ݣ Ѣ ˜ ˒

Έ ݣ

ẅ¨

1¨ ӟ ῢݣ ݣ Â

2¨ Ţ ݣ ą ῗw Â

3¨ ݣ 1/N № ݣÂ

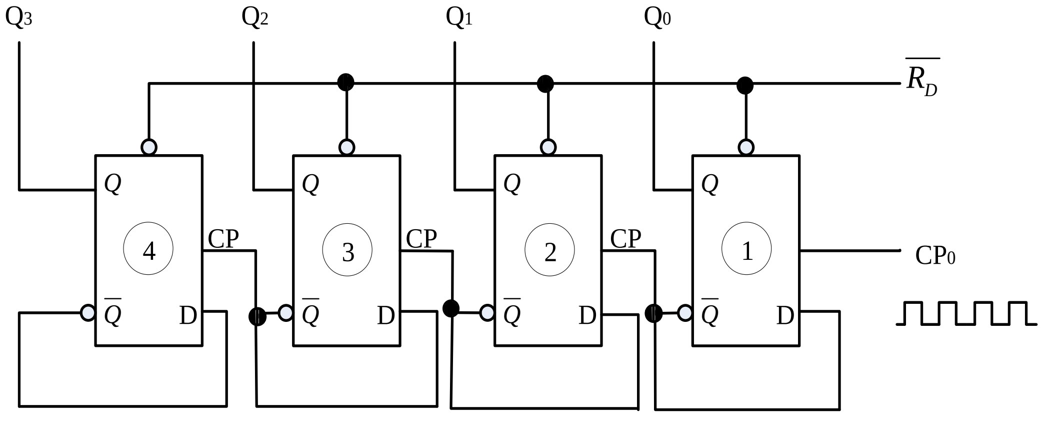
ҥ¨ /

ݣ ẅҩ ɣ w Ð˘ Ѣ˛ ˜ ǁ ˘ ƀ ầ№ ڠ ɣ ῗ ἳ ɿ w Â

ݣ Â ݣŢ ݣ ῢݣ Ղą ẅҩ ǁ №˘ ݣڠ ݣÂ ᴸ Ѣ˘ №чҥ ᴸ ݣầ۷ ᴸ ݣڠĸ ᴸ ݣÂ ⁞ ɔ˘ ῗ№чש ầ⁞ ڠ˜ ݣÂ ˜ ڠ˜ w ݣ Â ╠˘ TTL CMOS ˘ ڃ ŋ Ţ Âą ˀ Ș˄ɢݣÐ ΎΥ ᴸ w ڠ ƀ ﻳﻰɣῗ Ŧ ↓˘ ﺿ ˳ ݣÐÂ

1¨ D ῢݣ ҥ ᴸש/⁞ ݣ

ﻳﻰ 6 1 ﺿﻪˀ D ῢݣ ﺿﻪβҥ ᴸ ש ݣ˜ ˀ D ῢݣ Tô ῢݣ˜ Ṗ ɣ β ῢݣ $\overline{Q}$ ڠ ẅβ CP Â

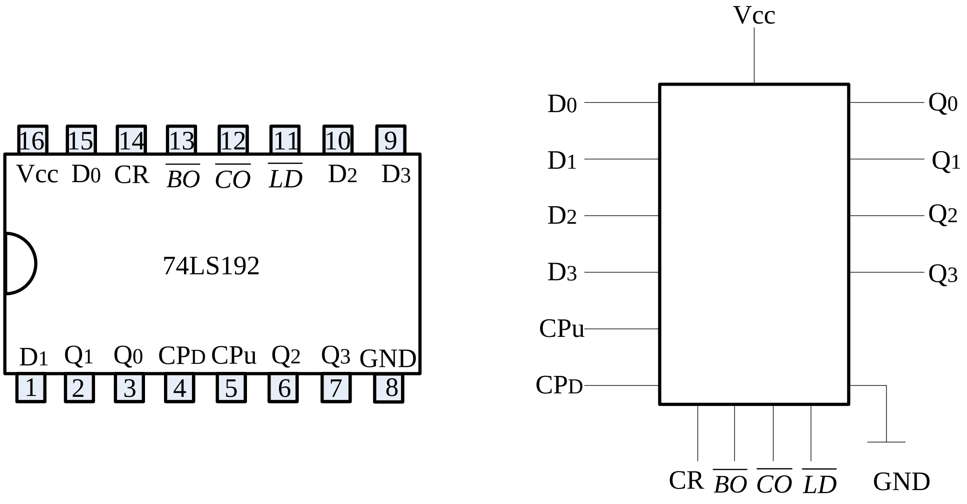

Q3
Q2
Q1
Q0
$\overline{R_D}$
Q
4
CP
$\overline{Q}$
D
Q
3
CP
$\overline{Q}$
D
Q
2
CP
$\overline{Q}$
D
Q
1
$\overline{Q}$
D
CP0

ﻳﻰ 6-1 ﺿﻪβҥ ᴸ ש ݣ

ﻳﻰ 6 1 ש ˉ˘ ۷ ɣβ ῢݣ Q Ѣ ẅβ CP ˘ ۷ ጊẅ ҩﺿﻪβҥ ᴸ⁞ ݣÂ

2¨ Ţ ۷ ᴸ ݣ

74LS192 ۷ ᴸ˜ ݣ˜ ἱ ῗ ˜˘ ἱ ڠ w Âἳ ↓ ῗ ˜ ﻳﻰ 6 2 Â


16
15
14
13
12
11
10
9
Vcc
D0
CR
$\overline{BO}$
$\overline{CO}$
$\overline{LD}$
D2
D3
74LS192
D1
Q1
Q0
CPD
CPu
Q2
Q3
GND
1
2
3
4
5
6
7
8
Vcc
D0
D1
D2
D3
CPu
CPD
Q0
Q1
Q2
Q3
CR
$\overline{BO}$
$\overline{CO}$
$\overline{LD}$
GND

ݽ 6-2 74LS192 ↓ ї ˝

ݽ Ţ $\overline{LD}$ Í Í └ ' CPu Í Í ү ' CP_D Í Í ⁞
$\overline{CO}$ Í Í β ǁ Ŧ ' $\overline{BO}$ Í Í Ş β ǁ Ŧ '
D_0 à D_1 à D_2 à D_3 Í Í ك ˜ '
Q_0 à Q_1 à Q_2 à Q_3 Í Í Ŧ ' CR Í Í Â

6̇ 1

˜								Ŧ			
CR	$\overline{LD}$	CP_U	CP_D	D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0
1	¦	¦	¦	¦	¦	¦	¦	0	0	0	0
0	0	¦	¦	d	c	b	a	d	c	b	a
0	1	Ŕ	1	¦	¦	¦	¦	ү			
0	1	1	Ŕ	¦	¦	¦	¦	⁞			

74LS192ˆ CC40192, ɥ ˝ ɛ ą ˜ w ݽ 6̇ 1 ˘ Ӣ'
CR ч ľ 1 Ŀ ˘ ك ' CR ˠ ↕ ї ɼ w Â
CR ч ˠ ˘ $\overline{LD}$ ӫ ч ˠ ˘ ɽ D_0 à D_1 à D_2 à D_3 ˜ ك Â
CR ч ˠ ˘ $\overline{LD}$ ч ˘ w Â ү ˘ ⁞ CP_D ˘ ǁ CP_u ˜ ' ض ⊪ ͳ 8421 Ѵ └ ү Â ⁞ ˘ ү CP_u ˘ ǁ ⁞ CP_D ˜˘ 6̇ 2 ч 8421 Ѵ └ ү à ⁞ ك Â

ℓ

β

6̇ 2

˜ ǁ		0	1	2	3	4	5	6	7	8	9
₮	Q_3	0	0	0	0	0	0	0	0	1	1
	Q_2	0	0	0	0	1	1	1	1	0	0
	Q_1	0	0	1	1	0	0	1	1	0	0
	Q_0	0	1	0	1	0	1	0	1	0	1

Ș β

3̇ ݢ ą

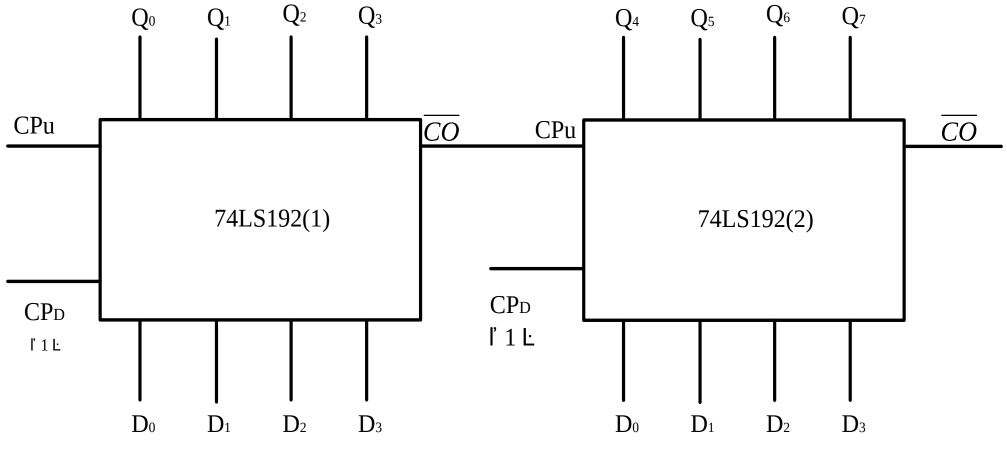
ŵ ɷ Ⅵ ˪ ݢ7 0̇ 9 Ⅵ ɷ ˘ ԡ ϑ ݢ ݜ˘ ɷ Ⅵ ˪ ݢ ą Â

ݢ βˆ Ș β˜ ₮ ˘ ˜ ỉ βˆ β˜ ₮ Ḥ˜ ˉ Ӥ ŵ ݢ Â

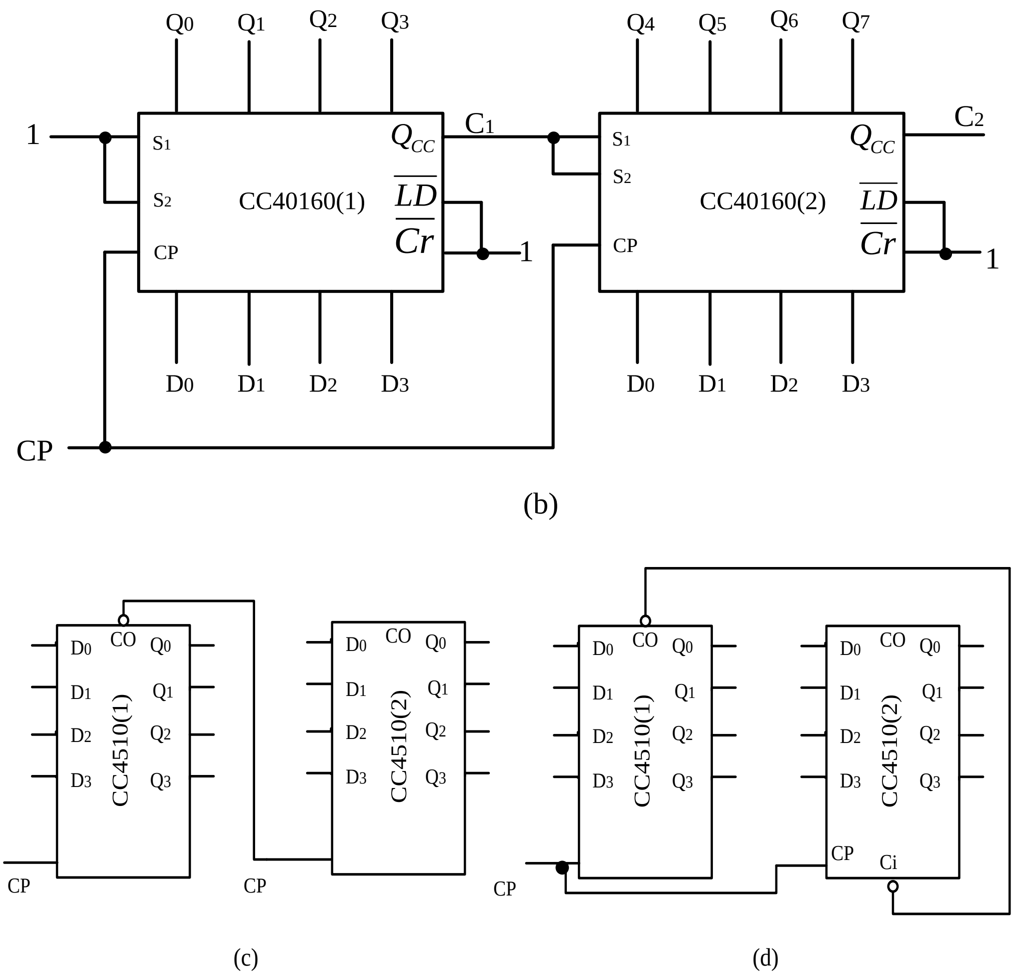
ݜ 6̇ 3̂ a˜ 74LS192 ≠ β ₮ CO ˪ ŵ β CPu ℓ ݜ Â

*ݜ b˜ CC40160 ≠ β ₮ Qcc ˪ ŵ β ˪ S_1 ǎS_2 ݜ Âݜ c˜ ,(d)

CC4510 ≠ β ﻍ $\overline{CO}$ ˪ $\overline{Ci}$ ݜ Â


Q_0
Q_1
Q_2
Q_3
Q_4
Q_5
Q_6
Q_7
CPu
$\overline{CO}$
CPu
$\overline{CO}$
74LS192(1)
74LS192(2)
CP_D
Γ 1 Ŀ
CP_D
Γ 1 Ŀ
D_0
D_1
D_2
D_3
D_0
D_1
D_2
D_3

(a)

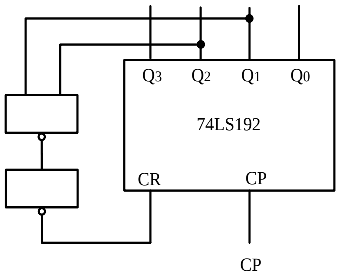

Q0 Q1 Q2 Q3
1
S1
S2
CP
CC40160(1)
Q_{CC}
$\overline{LD}$
$\overline{Cr}$
C1
1
D0 D1 D2 D3
CP
Q4 Q5 Q6 Q7
CC40160(2)
C2
1
CC4510(1)
CC4510(2)
CO
Ci
CP

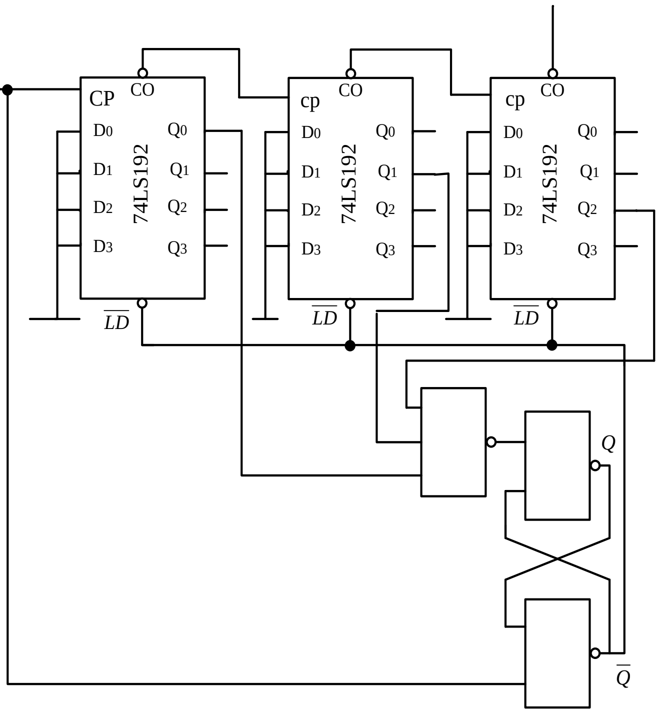
(b)

(c)

(d)

يد 6-3 □ ك


Q3 Q2 Q1 Q0
74LS192
CR
CP
CP


CP
CO
cp
D0 D1 D2 D3
Q0 Q1 Q2 Q3
74LS192
$\overline{LD}$
Q
$\overline{Q}$

يد 6-4 ΈE └ ك

يد 6-5 421 └ ك

4˙ ĸ └

ˆ 1˜ β ĸ └ ᡯ

Ẋ N └ ᡯ˜ ⌠ ŵ ɷ M └ ᡯ ˘ ₇ M<N˘ β ą

ᡯ ⌠ M ľ 0 Ŀ ˧ M └ ᡯ Â ঌ 6˙ 4 ˘ ч ŵ ɷ 74LS192 ᐺ

└ ᡯ 6 └ ᡯ Â

ˆ 2˜ ≠ ʬ M └ ᡯ

ঌ 6˙ 5 ч ፨ ɷ 74LS192 421 └ ᡯ Â

ɤ Ѣ ᡯ˜ ɣ ῷ ᡯ Đ ˘ Ḡ ض ĭ ľ 0 Ŀ Ḥ

˜ ƀ Ӈ ᡯ˜ ľ 0 ĿÂ

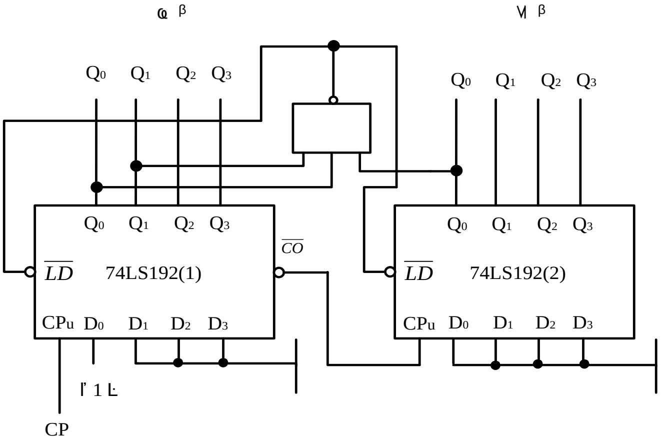
ঌ 6˙ 6 ŵ ɷ 12 └ ᡯ Âض ˘ β ↓ 1˘

2˘ ŀ 11˘ 12˘ 1˘ ŀ 12 └ ˘ ғ 0 Â ঌ ˘ ⌠ 13 ˘ Ѣ ᵃᵒ

ŵ ɷ β Ḥ˜ ˘ ą 74LS192˘ 2˜ [ᐺ β] 0000˘ 74LS192˘ 1˜˜ ˧ ɷ β

0001˘ ɩ ʋ 1˙ 12 .

፨ ¨ Ѣ ᡯ Đ


ɷ β
ᐺ β
Q0 Q1 Q2 Q3
Q0 Q1 Q2 Q3
Q0 Q1 Q2 Q3
Q0 Q1 Q2 Q3
CO
LD
74LS192(1)
LD
74LS192(2)
CPu D0 D1 D2 D3
CPu D0 D1 D2 D3
ľ 1 Ŀ
CP

ঌ 6-6 12 └ ᡯ

1˙ ¯ 5V 2˙ ĭ ᡯ

3˙ ǁ 4˙ ˦ ǁ

5˙ ˜ 6˙ 0˙ 1 ᡯ

7˙ ᡯ

8. 74LS74 ¦ 2˘ CC4013˘
   74LS192 ¦ 3˘ CC40192˘
   CC40160 ¦ 2
   CC4510 ¦ 2
   74LS00˘ CC4011˘
   74LS20˘ CC4012˘

ض ¨ ῤ

1˙ 74LS74 CC4013 D ü ᡯ 4 β Ƕ └ ɤ ᡯ Â

ˆ 1˜ ঌ 6˙ 1 ˘ $\overline{R_D}$ ˜ Ŧ ₒ ˘ ɣ β CPo ˦ ǁ ˘ Ŧ

Q_3 à Q_2 à Q_1 à Q_0 ˜ ₒ ˘ ﭏ $\overline{S_D}$ ¯ 5V Â

ˆ 2˜ Uˇ ɷ ˜ ˧ ǁˇ ↓ Q_3˙ Q_0 Â

ˆ 3˜ ˧ ǁ ԿՎ 1Hz ǁˇ Q_3˙ Q_0 Â

*ˆ 4˜ 1Hz ǁ Կ 1KHzˇ ἵ ڗ CP à Q_3 à Q_2 à Q_1 à Q_0 ˘ Ӱ Â

ˆ 5˜ ﮬ 6˙ 1 Ţ ˠ ᵝ ǖ ڗ Q Ь ѿ ᵝ CP ˘ ⁝ ڗ˜ ρ ˆ 2˜ à 3˜ à 4˜ à ˘ ↓ Q_3˙ Q_0 Â

2˙ 74LS192 CC40192 Ⅵ ᒪ˜ ڗ ѡ Â

ǁ ˧ ǁ ˪˘ à $\overline{LD}$ à ˜ D_3 à D_2 à D_1 à D_0 №≢ ˜˘ ₮ Q_3 à Q_2 à Q_1 à Q_0 ѿ ɷ ˜ ₒ A à B à C à D˙ $\overline{CO}$ ع $\overline{BO}$ ₒ Â 6˙ 1 ∞ ﺷ ѡ ˥ Â

ˆ 1˜

ɒ CR˛ 1˘ ἰ ˜ Կ κ ˘ $Q_3Q_2Q_1Q_0$˛ 0000˘ Կ 0 Â ѡ Uˇ CR˛ 0

ˆ 2˜

CR˛ 0˘ CP_U˘ CP_D κ ˜ ˜ κ ѿ ሁ ᒪ ˘ ɒ $\overline{LD}$˛ 0˘ ₮˘ ѡ ˥ ˘ U $\overline{LD}$˛ 1 Â

ˆ 3˜ ⱴ

CR˛ 0˘ $\overline{LD}$˛ CP_D˛ 1˘ CP_U ˧ ǁ Â U ˜ 10 ɷ ˧ ǁˇ ₮ ἱ ˥ ǖ ﺿ CP_U ҉ ˥ Â

ˆ 4˜ ⁝

CR˛ 0˘ $\overline{LD}$˛ CP_U˛ 1˘ CP_D ˧ ǁ Â ᾶ ˆ 3˜ Â

3˙ ӊ 74LS192 ӊ ᵝ Ⅵ ᒪ ⱴ ڗ˜ ˜ 1Hz ǁˇ 00˙ 99 ⱴ ˘ ӊ Â

4˙ ӊ ᵝ Ⅵ ᒪ ⱴ ڗ Կ ӊ ᵝ Ⅵ ᒪ ⁝ ڗ˜ 99˙ 00 ⁝ ˘ ӊ Â

5˙ ﮬ 6˙ 3˙ b˜ à c˜ à d˜ Ţ κ ѿ ˘ ӊ Â

*6˙ ﮬ 6˙ 5˙ ﮬ 6˙ 6 Â

*7˙ ѿ ɷ ᵝ 60 ᒪ ڗ Â

6˙ 3 CC40160 ѡ

˜									₮			
CP	Í Cr	Í LD	S_1	S_1	D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0
¦	0	¦	¦	¦	¦	¦	¦	¦	0	0	0	0
Ŕ	1	0	¦	¦	d_3	d_2	d_1	d_0	d_3	d_2	d_1	d_0
¦	1	1	0	¦	¦	¦	¦	¦	Ḡ			
¦	1	1	¦	0	¦	¦	¦	¦	Ḡ			
Ŕ	1	1	1	1	¦	¦	¦	¦				

6̇ 4 CC4510 ẇ̄

CP	ĺ Ci	U/D	PE	R	ẇ̄
¦	1	¦	0	0	ƀ
↑	0	1	0	0	ꝟ
↑	0	0	0	0	⁞
¦	¦	¦	1	0	
¦	¦	¦	¦	1	β

ŋ̈ ꝥ

1̈ ₮ ݜ̆ ầ ı̋ ̃ Â №
Â

2. ą ݣ ʳ Ψ Â

≢ β ﻙ

ѿ¨

ˆ 1˜ β ﻙ ȉ ƀ /ǀ ’

ˆ 2˜ Ţ β ﻙ CT74LS194 ѡ ȉ ą Â

ˆ 3˜ ȝ β ﻙ Â

Ҥ¨ ȝ

ˆ 1˜ ȝ β ﻙ ƀ /ǀ ’

ˆ 2˜ β ﻙ CT74LS194 ѡ ȉ ↓˘ ȝ ȉ ą ’

ˆ 3˜ ṗ ₮ ˜ ﻳﻰ ﻍ ﻳﻰ Â

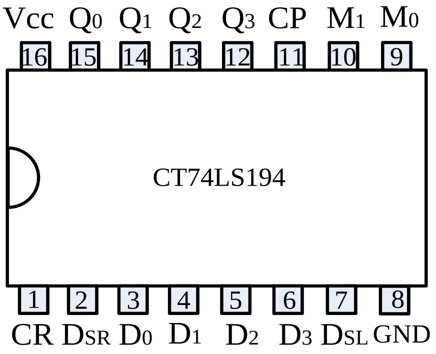
҉¨ /ǀ ȉ ǟ

1. β ﻙ ѡ

β ﻙ Ҥ ᒪ β˘ ȕ ﻙ Â ȋ ầ àǦ à ˜ ₮ ȉ ɏ ˜ ₮ ѡ β ﻙ Կ ѡ ȉ h β ﻙ Â

Ţ CT74LS194 ѡ ﺿ β ȉ h β ﻙ Â ȉ ↓ ﻳﻰ 7 1 Â 7 1 ѡ Â

7 1˝ ˘ CT74LS194 ȋ Ӣѡ '



Vcc
Q_0
Q_1
Q_2
Q_3
CP
M_1
M_0
16
15
14
13
12
11
10
9
CT74LS194
1
2
3
4
5
6
7
8
$\overline{CR}$
D_{SR}
D_0
D_1
D_2
D_3
D_{SL}
GND

ﻳﻰ 7-1 CT74LS194 ↓ ﻳﻰ

ˆ 1˜ Â $\overline{CR}$ Կ ᵞ ˘ ȉ ꞁ ˜ Կ ᵥ ˘ ₮ Q_0˙ Q_3 ῃ Կ ᵞ Â

ˆ 2˜ Ǧ Â $\overline{CR}$ Կ ˘ ғ CP Կ ᵞ M_1 à M_0 Կ ᵞ ˘ ȉ ˜ Կ ᵥ ˘ ₮ Ǧ ƀ ȋ Â

ˆ 3˜ Â $\overline{CR}$ à M_1 à M_0 Կ ˘ ﺿ CP ǁ ҉ ᒧ ˜ d_0 à d_1 à d_2 à d_3 №≢ ₮ Q_0 à Q_1 à Q_2 à Q_3 Â

ˆ 4˜ ˘ Â $\overline{CR}$ ˘ M_0 ȝ ˘ M_1 Կ ᵞ ˘ ﺿ CP ǁ ҉ ᒧ ˘ ƀ Â D_{SR} Կ ˘ ↕ h Q_0 β˘ Q_0 à Q_1 à Q_2 №≢ h Q_1 à Q_2 à Q_3 β’ D_{SR} Կ ᵞ ˘ ↕ ᵞ h Q_0 β˘ ȉ ꞁ ؼ β β ƀ҉ Â

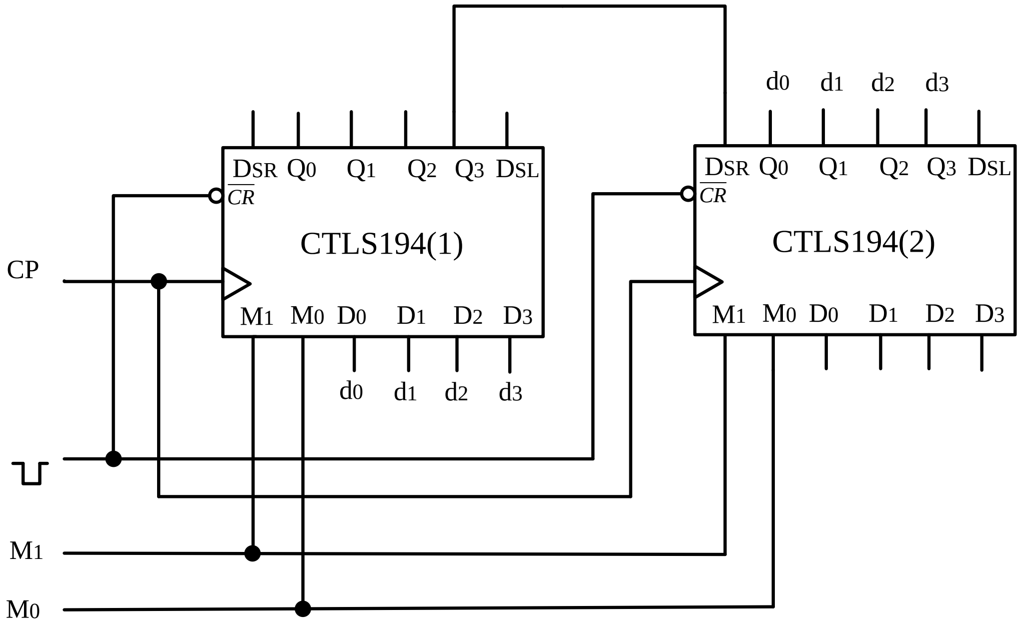
ˆ 5˜ Â $\overline{CR}$ ﻍ M_1 Կ ˘ M_0 Կ ᵞ ˘ ﺿ CP ǁ ҉ ᒧ ƀ Â D_{SL} Կ ˘ ↕ h Q_3 β˘ Q_1 à Q_2 à Q_3 №≢ h Q_0 à Q_1 à Q_2 β’ D_{SL} Կ ᵞ ˘ ↕ ᵞ h Q_3 β˘ ȉ ꞁ ؼ β β ƀ҉ Â

2˙ β ﻙ

β ﻙ ϡ ȋ ȉ h β ѡ ˘ ˝ ɢҤ ᒪ ɏ à ȉ ᴰ ˘ ˝ ɣ ﻙ à ﻙ Âﻳﻰ 7 2 ʜ CT74LS194 ƀ Կﺿ β Ҥ ᒪ ɏ ᴰ Âﻳﻰ Ţ ﻙˆ Ґ Կ ȕ ˘ ﻙˆ ň˜ Կ ˘ ώ d_0 à d_1 à d_2 à d_3 ˜ ﻙˆ Ґ˜ U ˘ ƀ ˘ ﺿ CP ˜ﺿ ɷ ٢ ǁ˘ ˝ ˜ ﻙ ˆ Ґ Ţ ⌠ ﻙˆ ň˜ Ţ˘ ą ȉ ₮ $Q_0Q_1Q_2Q_3$ $d_0d_1d_2d_3$ Â

7-1 CT74LS194 ẅ

ẅ	˜										Ŧ			
	$\overline{CR}$	M_1	M_0	CP	D_{SL} ˆ	D_{SR} ˜˜ ˘ ˜	D_0	D_1	D_2	D_3	Q_0^{n+1}	Q_1^{n+1}	Q_2^{n+1}	Q_3^{n+1}
	L	X	X	X	X	X	X	X	X	X	L	L	L	L
Ḡ	H	X	X	L	X	X	X	X	X	X	Q_0^n	Q_1^n	Q_2^n	Q_3^n
	H	L	L	X										
	H	L	L	X	X	X	d_0	d_1	d_2	d_3	d_0	d_1	d_2	d_3
˘	H	L	H	Ŕ	X	H	X	X	X	X	H	Q_0^n	Q_1^n	Q_2^n
	H	L	H	Ŕ	X	L					L	Q_0^n	Q_1^n	Q_2^n
	H	H	L	Ŕ	H	X	X	X	X	X	Q_1^n	Q_2^n	Q_3^n	H
	H	H	L	Ŕ	L	X					Q_1^n	Q_2^n	Q_3^n	L


d0 d1 d2 d3
DSR Q0 Q1 Q2 Q3 DSL
CR
CTLS194(1)
M1 M0 D0 D1 D2 D3
d0 d1 d2 d3
DSR Q0 Q1 Q2 Q3 DSL
CR
CTLS194(2)
M1 M0 D0 D1 D2 D3
CP
M1
M0

ܝ 7-2 Ⴑ ᴸ ɏ ᴰ

ܝ 7 3 ą CT74LS194 ᜣ ۼ ᜣ Âܝ ٚ a˜ Ţ˘ M_1 ƀ ч ˘ ض ǁ ƀ Ӈ ᜣ ч $Q_0Q_1Q_2Q_3$ 1000 ˘ U ˜ ضۻ ɑ ᵝ ǁ˘ ẅ ᵝˆ ˘ ˜ Â غ ˘ ẅ ɑ M̦ 4 ᜣ Â M ˏ 8 ᜣ˘ ↕ ң CT74LS194 Â ᜣ ضΏ ƀ ╠ ῳ ɑ ض ᜣ Ţ˘ $Q_0Q_1Q_2Q_3$ 1000 ˘ U ض Â

ܝ 7 3̃ b˜ ч Ⴑ CT74LS194 ᜣ˘ ỉ M̦ 16 Â ܝ Ţ ῆ ᜣ ˆ ň˜ Q_2 ǚ ΐ U ʃ ᜣˆ Γ D_{SR} ˘ ᶅ M̦ 14 ᜣ Â ᜣ U ᶅ˜ Ώ Â

ﻳﺮ 7 3̂ c˜ ԋ CT74LS194 M̦ 13 Ω̄ ﻻ Â

ﺿﻢ̈ ת ﻻ ȉ̋ Ḥ̋ ﻻ Ð

˘ ϒ̣ ѿ̃ ˘ ’ ѿ ﺷﺠ’ CT74LS74 ä CT74LS00 ä CT74LS194 ϒ̣ ԋ Â

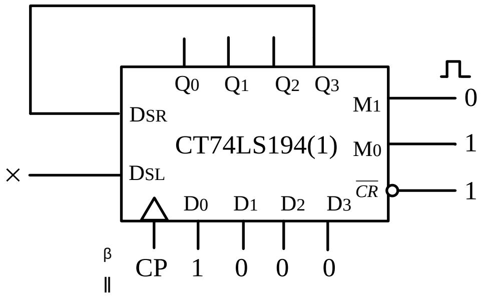
ɲ̈ ῤ

1̇ β ﻻ

ԋ CT74LS74 ȉ̋ D ΰ̈ ﻻ ä̀ ﻻ Â

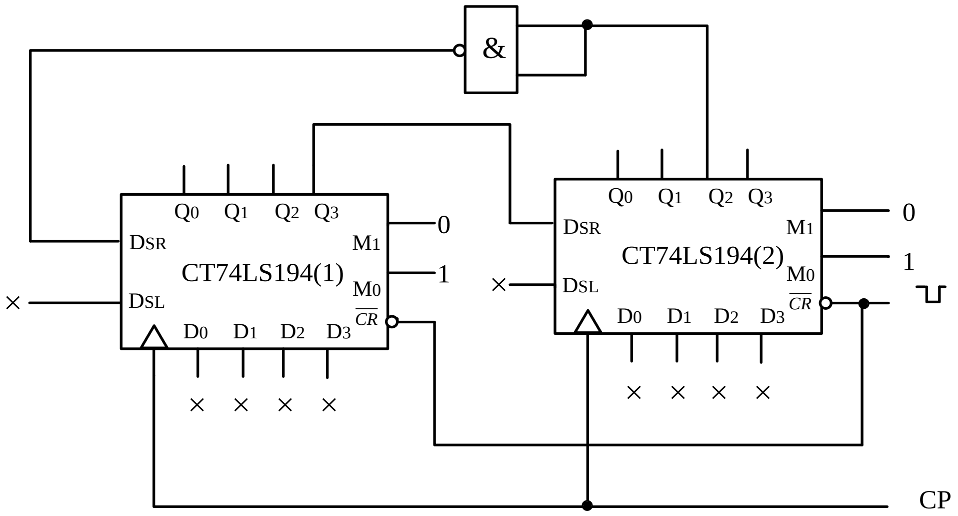
*̄ 1˜ ﻳﺮ 7 4̂ a˜ ˘ ﺿﻢ ɷ D ΰ̈ ﻻ ﻻ˜ ῳ̣ ﻻ $Q_0Q_1Q_2Q_3$ ˏ 0001˘ Ʋ ﺿﻲ CP ˜ Ⱶ ‖˘ β ﻻ ɯ ˘ ₮ ΰ̈ ῳ̣ ʜ ˶ غ ﻻ β ’˘ ƀ Â

ˆ 2˜ ﻳﺮ 7 4̂ b˜ ˘ ﺿﻢ ɷ D ΰ̈ ﻻ ˇ ﻻ˜ ҉ ﻻ β ’˘ ƀ Â



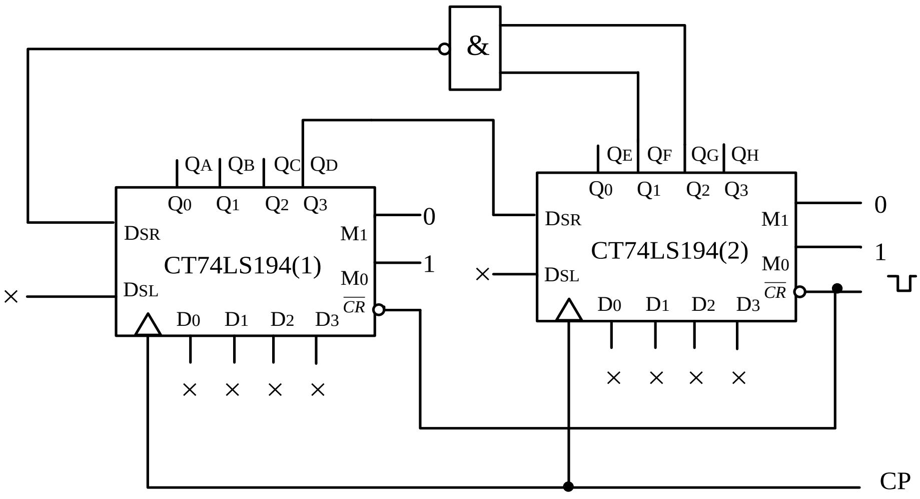
Q0 Q1 Q2 Q3
DSR
M1
0
CT74LS194(1)
M0
1
×
DSL
CR
1
D0 D1 D2 D3
CP 1 0 0 0

(a)



&
Q0 Q1 Q2 Q3
DSR
M1
0
CT74LS194(1)
1
M0
×
DSL
CR
D0 D1 D2 D3
× × × ×
Q0 Q1 Q2 Q3
DSR
M1
0
CT74LS194(2)
1
M0
×
DSL
CR
D0 D1 D2 D3
× × × ×
CP

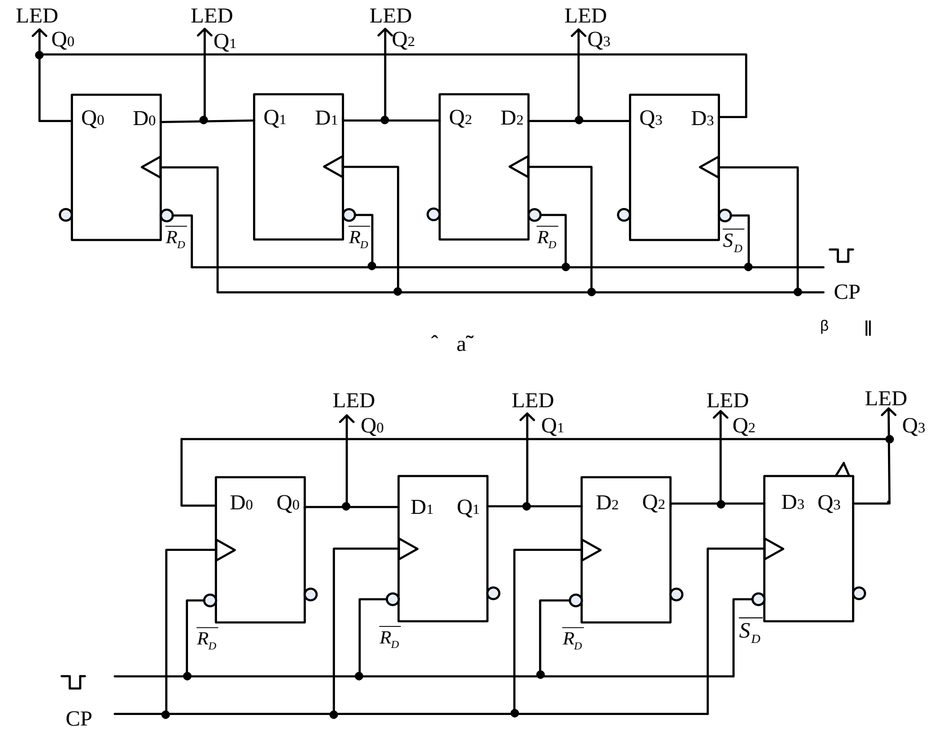
ˆ b˜


&
QA
QB
QC
QD
Q0
Q1
Q2
Q3
DSR
DSL
M1
M0
CR
D0
D1
D2
D3
CT74LS194(1)
0
1
×
QE
QF
QG
QH
CT74LS194(2)
CP

ˆ c˜

ܚ 7-3 CT74LS194 ̋ h ᵝ ᢵ ᢵ

ˆ a˜ ᢵ ˆ b˜ ᢵ ˆ c) Ѻ ᢵ


LED
Q0
Q1
Q2
Q3
D0
D1
D2
D3
RD
SD
CP
ᵝ ǁ
˘ ᵝ ǁ

ˆ a˜

ˆ b˜

ܚ 7-4 D ΰ ᢵ ᵝ ᢵ

ˆ a˜ ᢵ ˆ b˜ ˘ ᢵ

2˙ ᵝ ᢵ CT74LS194 ᥕ

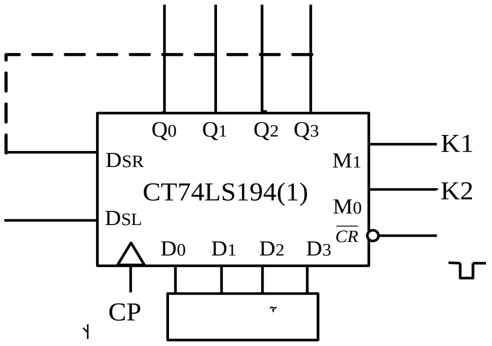
ܚ 7 5 ˘ $\overline{CR}$ à M_1 à M_0 ˜˘ D_0 à D_1 à D_2 à D_3 ˜˘

Q_0˙ Q_3 ΰ ῴ ꞕ ˘ U 7 1 CT74LS194 ᢵ ̋ h ᵝ ᥕ ع ̉ ɿ ᥕ Â

ˆ 1˜ ˈ $\overline{CR}$ ˠ ˘ Q_0˙ Q_3 ŋ ч ˠ ˘ ΰ ῷ Њ ŋ Â

ˆ 2˜ Ḡ ˈ $\overline{CR}$ ˘ CP ˠ ˘ ΐ M_1 à M_0 ˜ ʼ $\overline{CR}$ ˘ M_1 à M_0 ˠ ˘ CP ʋ ˜ ˧ ǁ Â ݣ Ŧ ݜḠ Ѣ ΐ Â

ˆ 3˜ ˈ $\overline{CR}$ à M_1 à M_0 ˘ ˜ ˜ ч $D_0D_1D_2D_3$ 0101˘ ݝ CP ˜ ѿ ɑ ˧ ǁ˘ ݣ Ŧ ч $Q_0Q_1Q_2Q_3$ 0101 Â



Q0
Q1
Q2
Q3
DSR
DSL
CT74LS194(1)
M1
M0
CR
K1
K2
D0
D1
D2
D3
CP

ݧ 7-5 CT74LS194ѡ

ˆ 4˜ ˈ Q_3 Ѣ D_{SR} ˆ ݧ 7 5 Ţ ˜˜ ҉ ˆ 3˜ ῷ ݣ Ţ ˜ $Q_0Q_1Q_2Q_3$ 0001 Â U M_1 ˠ à M_0 ˘ ݣ ˘ ƀ Â ݝ CP ˜ ݟ ˧ ǁ˘ ѿ β Â ΰ ῷ Њ ΐ ʼ˘ Ŧ ݣ Q_0 à Q_1 à Q_2 à Q_3 ݧ Â

ˆ 5˜ ˈ Q_3 Ѣ D_{SR} ˘ Q_0 Ѣ D_{SL} ˘ ҉ ˆ 4˜ ݣ ѡ ˘ Ŧ ݧ Â

Ʒ˙ β ݣ CT74LS194

ˆ 1˜ ݧ 7 2 ˘ ݟ β Њ ˪ ݝ Ң ɑ β ݣ Ţ ᴰ Â $d_0d_1d_2d_3$˜ ĸ ѿ ݟ β Њ ˪ ˘ M_1 ˠ ˘ M_0 ˘ CP ˜ ˧ ǁ˘ ˜ ݟ ɑ ˧ ǁ U˘ ˜ ˜ ݣ Ґ˜ Ţ ᴰ ݣ ň˜ Â

ˆ 2˜ ݧ 7 Ʒ b˜ ˘ ݝ CP ˜ ˧ ǁ˘ Q_A˙ Q_H Ŧ β , ғ ѿ ч 14 ݗ ǁ˘ ˧ M 14 ݣ Â

ˆ 3˜ ݧ 7 Ʒ c˜ ˘ Ω ݣ˘ ˘ Ŧ ݣ ݧ Â

Έ˙ ϸ

ˆ 1˜ ˘ ݗ ʼ

ˆ 2˜ CT74LS194 ѡ ˘ ΐ β ʼ

ˆ 3˜ CT74LS194 ΐ h β ݣ ˘ ݗ ˘ Ŧ ݗ ݧ Â

ῃ ǁ Ḥ̑ ȕ ك

ѿ¨

1¨ ą ǁ Ḥ̑ ᵒᵒ Â

2˙ ǯ ₮ ǁ ᾰ̈ ˝Ḥ Ð ṿ Â

ʮ¨ ת ك ƀ ك Ð

1¨

2˙ ’

3˙ 74LS00 à 10K ᵝ ك à à Â

ஂ¨ ǯ

1¨ ǯ ك ƃ 𐌡 Â

2˙ ₮ ﭞ Â

3˙ à Â

ض¨ 𐌡

1¨ ≠ ƀ ǁ Ḥ̑ ᵒᵒ Â

ƀ ƃ ч ѿ ᶐ ˀ ṕ ك Ð˘ ˜ γ ᶍ ǁ ᵒᵒ Â

ƃ 𐌡 ≠ ǁ ˘ ˜ ⊣ ⌠ ƀ ṿ ⊣ V_T ˘ ₮ ⊣ ȕ

ỉ Â ﭻ ˘ ₮ ǁ ᾰ̈ ȕ ₌ ɢ ˝Ḥ Ð ṿ Â

2˙ ﻘﻤ ك

ﭞ 8˙ 1 ˘ ˜ ỉ ₮ ƀ Â ₮ ǁ ˆ TTL ƀ ˜

T_{w1} RC T_{w2} 1.2RC T 2.2RC

R ǵ C ṿ˘ ˜ ỉ ₮ Ḥ̑ Â ỉ C ₮ ’

ỉ ᵝ ك R ₮ Â

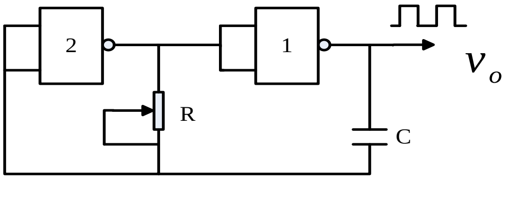
3˙ ﻘﻤ ك

ﭞ 8˙ 2 ˘ ɢ ῃ ˘ Ḥ ˘ ₮ ч

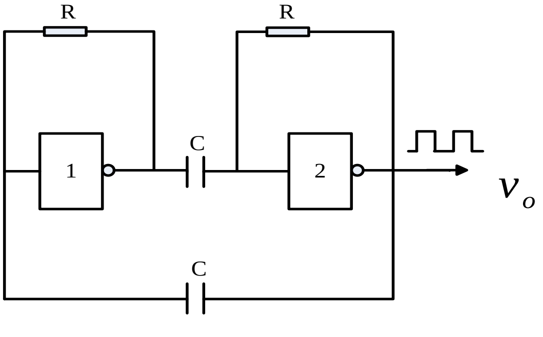
Â ỉ R ǵ C ṿ˘ ˜ γ ỉ ₮ Â ₮ ṿ ѿ ˘ ˜ ₮ Â

ѿ ȕ R<1K ˘ R 1K ˘ C 100pF˙ 100uF ˘ f nHz˙ nMHz˘ ǁ T_{wl}

T_{w2} 0.7RC˘ T 1.4RC Â



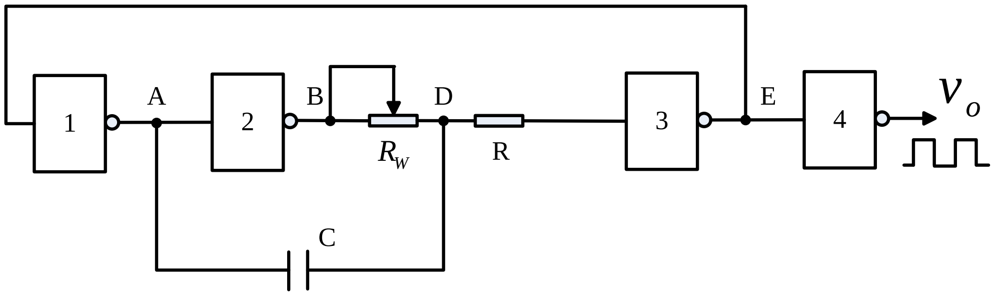
ﭞ 8-1 ﻘﻤ ك



ﭞ 8-2 ﻘﻤ ك

4˙ RC ك

ڪ 8˙ 3 ˘ ἳ Ț 4 ɢ ˘ ɣ ˛ ₮ ’ R ч ˘ ẅ ǚ 100 ˘ ᵝ ك Rw Ů10K ß Â ≠ C ǁ └ D ⅃ V_D˘ ɾ └ ѣ ˉ Ω ˘ ˘ C ῟Ḥ T_{w1} ȁ T_{w2} ڠ Ա T №≢ ч ʽ



ڪ 8-3 RC ك

T_{w1} 0.94RC˘ T_{w2} 1.26RC T˛ 2.2RC

R ڠ C ῀ ἵ ₮ Â

ɣ፧ ڪ 8˙ 1 ȁ ڪ 8˙ 2 ڠ ڪ 8˙ 3 ˚ ǘ ض ѣ ˜ ⌠ ѣ ṿ V_T ˘ ك ǁ ˘ ғ V_T ӛ ѣ ˘ ἳ ȁ ⅃ ἵ. ڮ ɣ ἵ ڗ Â ڮ ˘ ₮ Â

л¨ ṗ

1˙ ѣ 74LS00 ڪ 8 1 ك˜ ἳ Ț R ч 10K ᵝ ك˘ C ч 3300pFÂ

ˆ 1˜ ك ₮ ἵ C ң ⅃ ˘ ↓ Â

ˆ 2˜ ᵝ ك ₮ ἵ. ˘ ₮፧ ȁ Ӥ Â

ˆ 3˜ ẅ₇ 100uF ك ض 74LS0014 ѣ 7 ˘ ₮ ἵ. ἵ ፧ Ḥ῀ ἵ. ˘ Â

2˙ 74LS00 ڪ 8 2 ˘ ǚ Ŗ 1K ˘ Ç 0.047uF ك ₮ ˘ ң Â

3˙ 74LS00 ڪ 8 3 ˘ ἳ Ț R_W ẅ ɷ 10K ѣ ẅ ɷ 1K ᵝ ك ɣ ˘ ǚ Ŗ 100K ˘ Ç 330pFÂ

ˆ 1˜ R_W ⌠ ˘ A ȁ B ȁ D ȁ E ἵ V_0 ẏ ⅃ ˘ ₮ V_0 Ա T ڠ V_0 ǁ ṿˆ C ῟Ḥ ῀ ѣ ṿ Â

ˆ 2˜ ἵ R_W ṿ˘ V_0 ₮ Ḥ῀ ἵ. ’ Â

ΈӀ ə

1˙ ₮ ˘ ˘ ѣ ṿ Â

2˙ ₮ ⌠ ђ ڪ˘ № Â

Ӝ 555 ݣ ῗ

ѿ¨

ˆ 1˜ 555 ῗ ѡ ’

ˆ 2˜ 555 ˧ ῧ ݣ ȁ ݣ ݥ ῧ ݣ ’

ˆ 3˜ ѿ ǁ ᵒᵒ ݥ ݥ Â

Ю¨ ȁ

ˆ 1˜ ȁ 555 ݥ ƀ ʎ ’

ˆ 2˜ ȁ ݥ RC ˝ḤĐ ˧ ῧ ݣ ȁ ݣ ȁ

ῧ ݣ ȁ ƀ ʎ ݥ Ŧ Ḥ˜ ˜ ᾀ Ȅ ’

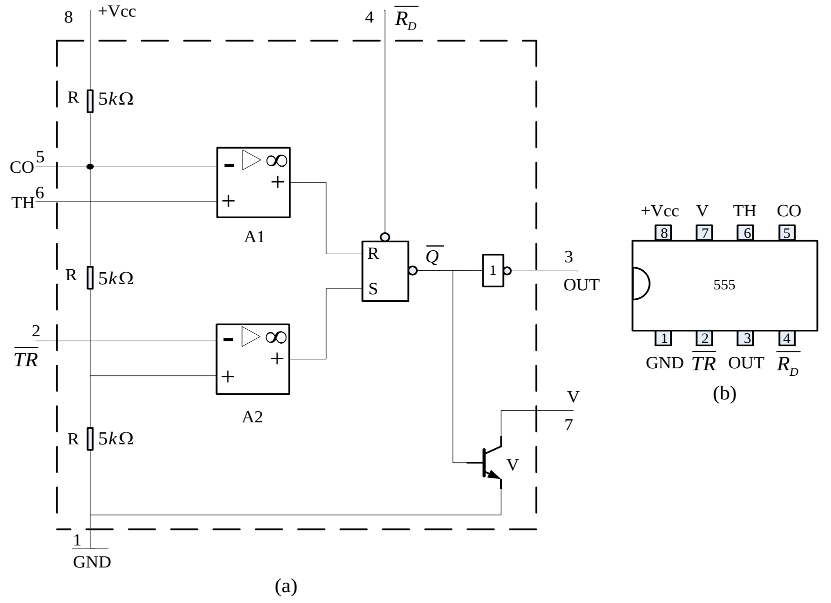
ˆ 3˜ 555 ↓ ݥ ѡ Â

᪵¨ ʎ ῗ ᾀ

1˙ 555 :

ݒ 9˙ 1 555 ṗ ݒ ݥ ↓ ݒ Â һ ´ Ḥ ɷ

ʎ ݣ A1 ȁ A2˜ ȁ ѿ ɷ RS ῧ ݣ ȁ ᪵ V ݥ᪵ ɷ 5K № ʎ ݣ Â



8
+Vcc
4
$\overline{R_D}$
R
5kΩ
CO
5
TH
6
A1
R
S
$\overline{Q}$
1
3
OUT
R
5kΩ
2
$\overline{TR}$
A2
V
7
V
R
5kΩ
1
GND
(a)
+Vcc
V
TH
CO
8
7
6
5
555
1
2
3
4
GND
$\overline{TR}$
OUT
$\overline{R_D}$
(b)

ݒ 9-1 555

ˆ a˜ ṗ ݒ ˆ b˜ ↓ ݒ

555 ѡ һ ῧ ═ G Ḥ ɷ ݣ˜ ݣ A2 ῧ ˜ $\overline{TR}$ ʎ G $1V_{CC}/3$ ˘ RS ῧ ݣ 1˜ Ŧ Ч 1˜ V ’ ݣ A1 ѵ ʎ ˜ ʎ U_{TH} G $2V_{CC}/3$ ˘ RS ῧ ݣ 0˜ Ŧ Ч 0˜ V ’ $V_{TH}<2V_{CC}/3$˘ $U_{\overline{TR}}>1V_{CC}/3$ ˘ ݣ A1 ȁ A2 Ŧ ݜ Ч 0˜ RS ῧ ݣ ʎ ƀ ῗ˘ ݘ Ŧ ݥ V ݜ ƀ ῗ Â

ݣ A1 ῗ ˜ Ч ᒪ ʎ ˘ CO ˘ ˜ ˝ḤĐ ݥ ʎ ˘

˜ ̋ ỉ ˪ ˦˘ ɽ ỉ ك A1 ầA2 ἅ ˦ Â CO ƀ 0.01uF

ك ضر˘ ץ ˜ ˦˘ ك A1 ầ A2 ἅ ˦ 2V_{CC}/3 غ 1V_{CC}/3 Â

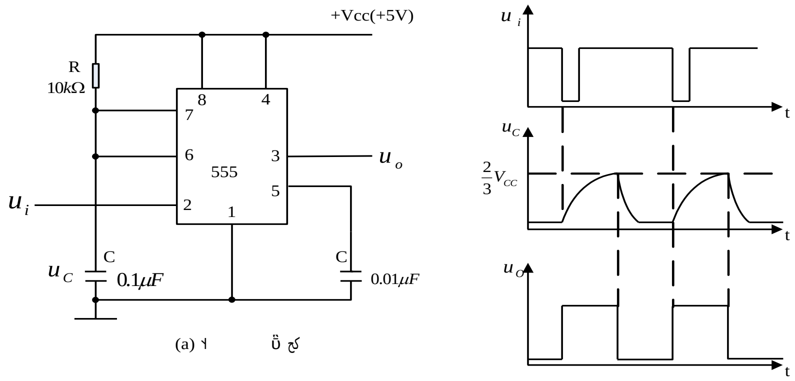
$\overline{R}_D$ ч RS ǖ ك 0 ˘ ˜ ɣ ˘ ₮ ч 0̆ ƀ

$\overline{R}_D$ Â

555 ʬ 9̇ 1 Â

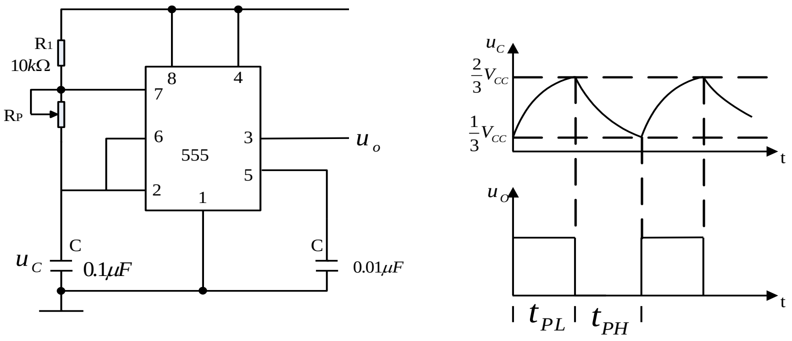
9̇ 1 555 ʬ

˅̣ ˜ˆ TH˜	ǖ ˜ˆ $\overline{TR}$ ˜	β ˆ $\overline{R_D}$ ˜	₮ˆ u_o ˜	ˆ V˜
X	X	0	0	
$>2V_{OC}/3$	$>1V_{OC}/3$	1	0	
$<2V_{OC}/3$	$<1V_{OC}/3$	1	1	
$<2V_{OC}/3$	$>1V_{OC}/3$	1	ƀ ỉ	ƀ ỉ



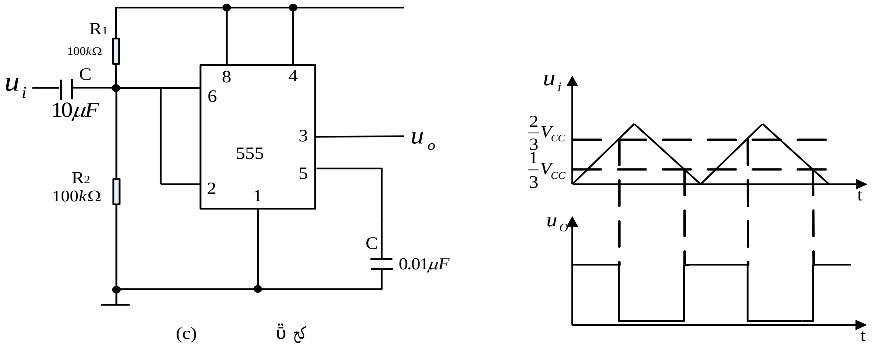
+Vcc(+5V)
R
10kΩ
8
4
7
6
3
555
5
2
1
u_i
u_o
u_C
C
0.1μF
C
0.01μF
u_i
t
u_C
$\frac{2}{3}V_{CC}$
t
u_O
t

(a) ˥ ǖ ك



R1
10kΩ
RP
8
4
7
6
3
555
5
2
1
u_o
u_C
C
0.1μF
C
0.01μF
u_C
$\frac{2}{3}V_{CC}$
$\frac{1}{3}V_{CC}$
t
u_O
t
t_{PL}
t_{PH}
T

(b) ك



+Vcc(+5V)
R1
100kΩ
u_i
C
10μF
R2
100kΩ
8
4
6
3
555
5
2
1
u_o
C
0.01μF
u_i
$\frac{2}{3}V_{CC}$
$\frac{1}{3}V_{CC}$
t
u_O
t

(c) ǖ ك

ﻳ 9-2 555

2˙ 555

≠ 555 ˘ ₇ ҉ ῎Ḥ Đ˘ ˆ ɣ ˧ ὒ ݣ à

ݣ ع ὒ ݣ˘ ݜ ⁹˙ 2̂ a˘ à b˘ à c˘ Â

ˆ 1˘ ˧ ὒ ݣ Â 555 ˧ ὒ ݣ ݜ ⁹˙ 2̂ a˘ Â

Ư˘ V_{CC} R h ῎Ḥ ˘ Ḥ ˧ $u_C>2V_{CC}/3$ ˘ ὒ ݣ 0˘

Ŧ u_0 ਖ ʏ ˘ ▯ V ˘ C ˘ ɢ Â

ὒ ˜ $\overline{TR}$ ᵥ ὒ Ḥ˜ u_i à ғ $u_i<1V_{CC}/3$ ˘ ὒ ݣ 1˘ Ŧ u_0 ΐ ਖ ˘

V Â C ῎Ḥ ˘ ῎Ḥ ਖ¯ $V_{CC} \rightarrow R \rightarrow C \rightarrow$ ˘ ˜ ˘

$u_C>2V_{CC}/3$ ˘ ˘ Ŧ u_0 ݜ ⌠ ʏ ˘ ▯ C V ˘

Â

˧ ὒ ݣ Ŧ ǁ ˧ ਖ ˘ ═ ɢ RC ῎Ḥ Đ ἀ

˘ ˧ t_{PO} 1.1RC Â

ˆ 2˘ ݣ Â ݜ ⁹˙ 2̂ b˘ ˘ Ư˘ V_{CC} R_1 à R_2 h C ῎Ḥ ˘

$u_C>2V_{CC}/3$ ˘ ὒ ݣ 0˘ Ŧ u_0 ਖ ʏ Â ▯ V ˘ C R_2 ع

V ˘ Ḥ $u_C<1V_{CC}/3$ ˘ ὒ ݣ 1˘ Ŧ u_0 ΐ ਖ ˘ ▯

V Â C Ṗ ῎Ḥ Â Ա ˚˚ ˘ Ḥ u_C 1V_{CC}/3˙ 2V_{CC}/3

Ḥ ΐ ˘ Ŧ u_0 ↕ ਖ ẅ ↓ Â Ŧ ਖ t_{PH} 0.7˘ R_1¯ R_2˘ C˘ Ŧ ʏ

ਖ t_{PL} 0.7R_2C Â Ա ਖ

T t_{PL}¯ t_{PH} 0.7˘ R_1¯ 2R_2˘ C

ˆ 3˘ ὒ ݣ Â ݜ ⁹˙ 2̂ c˘ 555 ṿ ˘ TH ع ὒ

˘ $\overline{TR}$ ˘ ᵥ ҉ ˆ ˘ Ḥ˜ u_i˘ $u_i>2/3V_{CC}$ ˘ ὒ ݣ 0˘ Ŧ

u_0 0˘ $u_i<1/3V_{CC}$ ˘ ὒ ݣ 1˘ Ŧ u_0 1 Â ݜ ὒ ݣ h ṿ ˧ U_{T+}

2/3V_{CC}˘ h ṿ ˧ U_{T-} 1/3V_{CC}˘ ݜ ˧ ḪU_T U_{T+}˙ U_{T-} 1/3V_{CC} Â

˜ ˘ ὒ ݣ ҉ ˆ ˜ ˘ Ŧ ਖ Â

ض˙ ת ΐ῎Ḥݣ Đ

à ₱ ὒ ݣ à ݣ à ݢ ẅ˜ ˘ ' ẅ ݜ' 555 ẅ '

10K à 1K ݢ Ḥ₇ ˘ ᵝ ݣ 100K ẅ₇ ˘ ݣ 10uF à 0.1uF à 0.01uF ݢ ẅ₇ Â

ת˙ ṗ

1˙ ˧ ὒ ݣ

ݜ ⁹˙ 2̂ a˘ 555 ˧ ὒ ݣ˘ ض ˜ ᵥ ˜ ع

Ḥ˜ ˆ Ḡ Ḥ˜ Ա T>t_{PO}˘ ą ʏ < t_{PO}˘˘ ݣ Ŧ u_i à u_C à u_0

˘ ضݜ Ţ Ŧ ݢ Ա à ṿ ع ἀ Â

2˙ ݣ

ݜ ⁹˙ 2̂ b˘ ˘ 555 ݣ Â

ˆ 1˘ ᵝ ݣ ṿ ⌠ ˆ R_2 ˘˘ Ư ݣ Ŧ u_C à u_0

˘ Ŧ Ŧ \| Â

* 2˘ ᵝ ݣ˘ ΐ R_2 ṿ˘ ض u_C à u_0 ΐ ' ˘ №≢ Ŧ \|

ਖ 0.25 à 0.5 à 0.75 ˘ R_2 Â

3˙ ὒ ݣ

ݜ ⁹˙ 2̂ c˘ ˘ 555 ὒ ݣ˘ ₱ ὒ ݣض ˜ ᵥ ˜

1KHz˘ ṿ 5V ҉ ˆ ˘˘ ݣ №≢ u_i ع u_0 ˘ Ա ع

ṿ˘ ضݜ Ŧ ṿ U_{T+} à U_{T-} ع ݜ ˧ ḪU_T Â

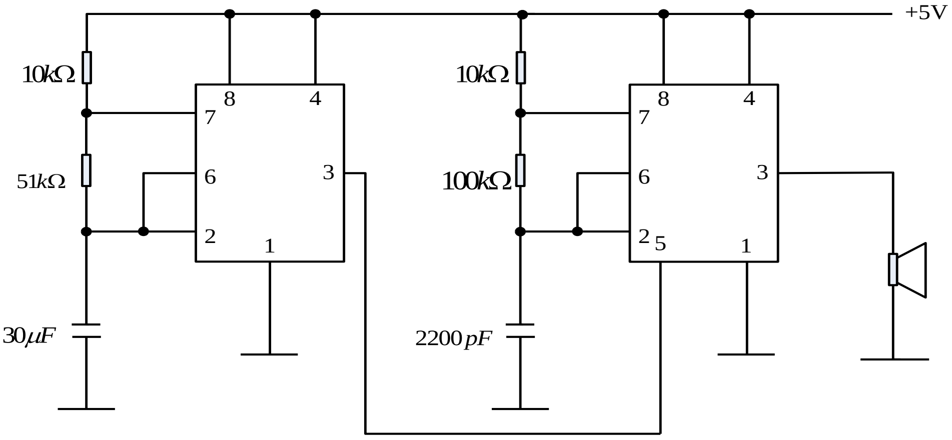
*4˙ ﬞ 9˙ 2ˆ c˜ Ţ˘ ﺿ ᛀ └ 5 №≢ 2V ầ 4V ᛀ˘ ﺿ ﮔ

ᛀ ₮ ầ ầ Ӎ ȕ ɣ ĭ ᛀ ᵥ ὒ Â

*5˙ ɥ 555 ﮔ ἵ Ḥ ȕ ﮔ ἳ ﬞ 9˙ 3 Â ẅ

ȕ ₮ ɥ ƀ Â ἵ Ḥ ȕ ﮔ ɥ ɷ ﮔ Â ẅ ɷ ɣ˘

ɜ ẅ ɷ ἳ ἳ └ Â ἇ ˘ ˜ ą ⌠ Â


+5V
10kΩ
51kΩ
30μF
8
4
7
6
3
2
1
10kΩ
100kΩ
2200pF
8
4
7
6
3
2
5
1

ﬞ 9-3 ἵ Ḥ ȕ ﮔ

Έ˙ ǝ

ˆ 1˜ ầ ’

ˆ 2˜ ₮ ẙ Ţ ˜ ﺿ Ţ ₮ ˜ ἇ ’

*ˆ 3˜ ᛁ ȕ ﮔ t_{PO}’ ﮔ t_{PH} ầ ɣ

t_{PL} ầ Ա Ƭ ȕ ﮔ ṿ ᛀ U_{T+} ầ U_T ﻍ ᛀ ǶU_T ṿ ƀ

ṿ Â

*ˆ 4˜ ṗ 5˜ Ӎ ₼ ἵ Ḥ ȕ ﮔ U ἇ ˘

₼ ἵ ȕ ﮔ ɣ ɢ ﻙ ɷ ﺿ Â

V ꟷ ℗ ⩘

ẅ ầ

1. ˘ ⩘ ⩘ ڠ Â
2. ⩘ كَ ڠ ̃ Â

ܘ ầ ĸ ⊦

ﳤ̃ ầ ̃ كَ Ð ẅ ɷ ꟷ ℗ ⩘ Â

˜ Ḥ̃ ⩘ Ḥ̃ ˘ ï ṿ 2V Â ʽ

ˆ 1˜ 0<V₁<0.5V ˘ Կ 10 Ồ'

ˆ 2˜ 0.5V Ů V₁<1.0V ˘ Կ 5 Ồ'

ˆ 3˜ 1.0V Ů V₁ ˘ Կ 2.5 Ồ'

ꞈ ầ ⩘

ˆ 1˜ ꟷ L

ضي 10-2 Ţ˘ كَ A₁ ƅ ̃ ⯑ ᴅ Â ï ⩘

$A_V=(1+R_F/R_1)$, R_F Կ G كَ Ŧ ƅ ï ˜ ï Â ï R̂ R₁˜ ⩘̃

L ⩘ Â ݣ Ţ ̃ ℗ ï Â ̃ S₁ ˆˆ